Parole Officer Effects on Reentry: Support, Deterrence, and Bias

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April 17, 2026

Abstract:

Little is known about the effects of the parolee-officer relationship. In this work, I use the quasi-random assignment of paroled individuals to parole officers in Pennsylvania to evaluate the effects of individual officers on reentry success. I find that individual officers affect parole violations and officer-documented employment, but not tax record-documented employment or arrests. Based on responses from a parole officer survey, officers that prioritize interpersonal support reduce parole violations relative to officers that prioritize structural support or deterrence. At the aggregate level, I find significant outgroup disparities – assignment to officers of a different race or sex increases arrests and decreases employment. These disparities are observed at a similar rate across parolee-officer pairings, suggesting they are due to systemic or societal biases that impact parolee-officer relationships as opposed to specific officers.

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1. Introduction

In the United States, one out of every 50 adults lives under correctional supervision. In 2019, 1.4 million individuals were incarcerated in State and Federal prisons, 0.9 million were living in the community under parole supervision, and 3.5 million were living in the community under probation supervision (Carson 2020; Oudekerk & Kaeble 2021; U.S. Census Bureau 2019). In addition, outcomes for supervised individuals are exceedingly poor. For incarcerated individuals, upon release from prison 71% are re-arrested within five years, 46% return to prison within five years (Durose & Antenangeli 2021), and 55% are unemployed eight months after release (Visher et al. 2008). For individuals released from prison early under parole supervision, recidivism rates are higher than for individuals released without supervision – in Pennsylvania in 2016 the three-year recidivism rate for individuals released to parole was 66.2% relative to 56.8% for individuals released without supervision at the end of their convicted sentence (Pennsylvania Department of Corrections, 2022). Finally, individuals under both probation and parole supervision realize poor outcomes – in 2019 31.1% and 36.9% of individuals who exited probation and parole supervision, respectively, exited unsuccessfully for reasons such as incarceration (for new arrests or broken supervisory conditions) or absconding (Oudekerk & Kaeble 2021).

Given the size of the U.S. correctional system and poor outcomes for supervised individuals, policy makers need clear evidence about the effects of community supervision to motivate decisions about how best to supervise and facilitate reentry success. Not surprisingly – given these poor outcomes – relatively little is known about the effects of community supervision (Doleac & LaForest 2022). As community supervision policy varies widely across states and municipalities in the United States (Phelps & Curry 2017), a better understanding of which policies and procedures are and are not effective, and why, is needed to improve the efficacy of community supervision programs.

One of the main components of community supervision is one's assigned probation or parole officer. Officers work with individuals assigned to probation or parole as both case workers (to facilitate reentry into the community) and law enforcement officers (to enforce supervisory conditions) (Chamberlain et al. 2017). Over many meetings, officers provide support to supervised individuals, build rapport with them, connect them with community programs and resources, and convey a clear message of deterrence regarding future criminal behavior and violations of supervisory conditions. When overseeing community supervision, officers have a wide degree of

discretion in how they build relationships with individuals, structure supervision, and whether to incarcerate when individuals break conditions – decisions that can have a large and lasting impact on a supervised individual's future.

This work explores the impact of parole officers on community success, for which little research exists. First, it investigates the extent to which one's parole officer affects success. That is, do certain officers improve outcomes for the individuals they are assigned, relative to other officers? Next, it investigates why certain officers affect post-release success. That is, are outcomes better when officers focus on interpersonal support, structural support, or deterrence? Finally, it investigates how demographic differences between officers and paroled individuals affect success. That is, does being assigned an officer of a different race or sex impact one's outcomes, and is this effect driven by particular officers?

This work investigates the impact of parole officers using individual-level data on the Pennsylvania parole population between 2006 and 2020. It estimates the causal effects of parole officers by leveraging the quasi-random assignment of paroled individuals to parole officers. Specifically, in Pennsylvania paroled individuals are first assigned to parole officer units based on their proposed residency location and required special needs (such as sexual offender needs and domestic violence offender needs). Within each parole officer unit, individuals are assigned to parole officers based on current officer caseload sizes. As such, similar paroled individuals assigned to the same officer unit are assigned to different officers based on slight differences in assignment dates.

I find that, first, individual officers have little relative effect on paroled individuals' future arrests, and at most a modest effect on major parole violations that lead to reincarceration. However, individual officers have a large effect on paroled individuals' future minor parole violations, that result in written warnings, additional parole conditions, and stays in non-secure facilities. Specifically, a one standard deviation increase in officer expected effect on minor parole violations increases the chance a paroled individual receives a minor violation in the first year after release by 14%. Finally, individual officers have a substantial effect on parole officer-documented employment but little effect on tax record-documented employment (from employer unemployment compensation quarterly tax reports). Taken together, these results suggest that individual officers primarily impact outcomes through variation in documentation and sanctioning decisions, as opposed to directly affecting paroled individual behavior.

Second, I leverage the results of a parole officer survey to investigate whether officer effects on outcomes are driven by prioritizing interpersonal support, structural support, or deterrence. I find that officers that focus on providing interpersonal support – noting that they care about paroled individuals, try to learn about them, and prioritize building good relationships and rapport – reduce minor parole violations. Officers that focus on skill-building and structural supports – noting that they work with paroled individuals to build pro-social behavior and focus on connecting them with community programs – have little observable effect on post-release outcomes. Officers that focus on providing reentry support relative to deterrence slightly increase minor parole violations. Overall, individuals assigned to officers that focus on interpersonal support have slightly better outcomes than individuals assigned to officers that focus on structural support or deterrence. Though these results may be driven by differences in officer reporting decisions, reductions in minor violations are still beneficial, as they reduce the burden of new parole conditions (added in response to violations) without negatively impacting future recidivism or employment (LaForest-Tucker, 2026). The results imply that employing more officers who focus on interpersonal support would improve outcomes by reducing parole violations. However, it is unknown whether these effects would be realized by training officers to focus on interpersonal support or, if effects are caused by inherent characteristics of officers that choose to focus on interpersonal support, would only be realized by hiring more officers who are interested in prioritizing interpersonal support.

While I find little relative effect of individual parole officers, I find significant aggregate-level outgroup effects for parolee-officer pairings, across all outcomes. Specifically, being assigned to an officer of a different racial background leads to an increase in future recidivism of 19%, driven by a 25% increase in future arrests, with suggestive evidence of a slight increase in parole violations and a slight decrease in employment. Being assigned to an officer of a different sex leads to a decrease in employment of 10% (for both officer-documented and tax record-documented employment measures), with suggestive evidence of an increase in arrests and parole violations. These effects are largely driven by outgroup disparities for female paroled individuals. Further analysis finds that these disparities are observed at a similar rate across parolee-officer relationships, rejecting a hypothesis that they are driven by a subset of potentially "biased" parole officers. Instead, the results support a hypothesis that the disparities are driven by structural and societal biases that impact the relationships paroled individuals and parole officers are able to

jointly construct. While they motive a conclusion that these disparities may only be fixed at the societal level, increasing officer awareness of these disparities could potentially help to reduce them.

In addition, this work provides an analysis of the employment experiences of paroled individuals, for which little prior knowledge exists. It combines parole officer-documented employment data – likely to be embellished and overreported – with tax record-documented employment, industry, and wage data – likely to be underreported. By combining these measures, this work bounds these reporting biases and provides a picture of the overall employment experience of individuals under parole supervision – finding an average employment rate of around 40% one year after the start of supervision.

Overall, little research exists on the effects of community supervision (Doleac & LaForest 2022). Of the small set of existing studies, most conclude that more intensive community supervision has little corrective impact. First, several randomized controlled trials find no effect of intensive probation programs relative to standard programs (Petersilia & Turner 1993; Lane et al. 2005; Henneguelle et al. 2010; Hyatt & Barnes 2014; Barnes et al. 2012). Recent work by Banan (2022) compares the effects of release to parole relative to release without supervision (using a regression discontinuity design in North Carolina) and finds that parole supervision leads to a short-term increase in parole violations that offsets a slight decrease in new crimes, with no long-term effects. Sakoda (2024) and Brinkman et al. (2025) find similar but slightly worse results – that additional time in the community under supervision leads to an increase in reimprisonment with no impact on reoffending (using a difference-in-differences design in Kansas and a policy change in Illinois, respectively).

Regarding the components of community supervision, Lee (2023) finds a negative effect of halfway house residency (using a randomized case worker design in Iowa) and Georgiou (2014) finds no effect of increased parole supervision intensity (using a regression discontinuity design in Washington). Rose (2021) finds that incarcerating individuals for probation violations has little deterrent effect but does increase racial disparities, with sanctions for failing to pay fines and fees having a particularly large disparate impact (using a policy change in North Carolina). Finally, LaForest-Tucker (2026) identifies the effects of parole supervision intensity and special conditions such as curfew, placement in a halfway house, or enrollment in a drug treatment program (using a regression discontinuity design and a randomized parole interviewer design in Pennsylvania,

respectively). He finds that, along most margins, increased supervision leads to additional parole violations with little effect on future arrests or employment.

Only one prior causal study exists on the effects of parole officers. Andersen and Wildeman (2015) find that officers in Denmark have a relative effect on dependency on public benefits and recidivism, but not employment (using a randomized parole officer design in Copenhagen).[2] My findings expand upon this study, and the sparce but unpromising nascent literature on the effects of community supervision, finding that parole officers in Pennsylvania impact parole violations (with officers that focus on interpersonal support reducing violations) but have little relative effect on arrests and employment. They motivate a result that increasing the number of officers who focus on interpersonal relationships would improve paroled individuals' outcomes, though differences in parole officer approaches are not a panacea that can fix the scale of the poor post-release outcomes that occur for paroled individuals.

A separate set of research focuses on the effects of racial bias, gender bias, and outgroup disparities. In the criminal justice setting, existing research largely focuses on how racial bias impacts police officer and judge decisions, in one-off interactions with potential offenders. For example, Arnold et al. (2018) find evidence of racial bias in judge bail decisions (using a randomized judge design in Miami and Philadelphia). Weisburst (2024) finds racial disparities in officer arrest propensities (using a randomized officer design in Dallas), and Gonclaves & Mello (2021) find that a subset of officers are less likely to provide discounts on speeding tickets to minorities, relative to white individuals (using a randomized officer design in Florida). Separately, Anwar & Fang (2006) find no evidence of extensive racial prejudice, but cannot reject a hypothesis of relative racial prejudice, for police motor vehicle searches (also using a randomized officer design in Florida).[3]

My work differs from this previous work in the criminal justice setting by investigating bias and outgroup disparities in a repeated interaction setting, as opposed to a single interaction

[2] All other existing studies on the effects of parole officers are correlational or qualitative. Bares & Mowen (2019) conduct a matching analysis using a panel data survey and find a correlation between officers that provide professional support and decreased recidivism, but no correlation between interpersonal support and recidivism. Morash et al. (2016) find no effect of individual officers on female individuals, using a matching analysis. Finally, Chamberlin et al. (2018) leverage a survey of paroled individuals and find that those that mentioned having better relationships with their parole officers had better parole outcomes.

[3] Kleinberg et al (2018) investigate the extent to which risk score algorithms can exacerbate or reduce racial biases. They show that including racial demographic details in risk score algorithms can produce more equitable and efficient recommendations than algorithms that omit racial demographic details in an attempt to be race-blind.

one-off police officer or judge decision. Here, parole officers and paroled individuals meet and interact many times over many months, building relationships. In the repeated interaction setting, observed outgroup outcome disparities could be driven by parole officer bias, paroled individual bias, or other societal biases that impact the tenor and quality of parolee-officer relationships. As such, this analysis shares many similarities with research on same-race and same-sex teacher effects in the education setting. In that setting, a wide variety of literature shows increased academic achievement when students are assigned to teachers of the same race (Dee, 2004; Fairlie et al., 2014; Egalite et al., 2015; Gershenson et al., 2022) and sex (Carrell et al., 2010; Gong et al., 2018), reduced absenteeism and suspensions (Lindsay and Hart, 2017; Holt and Gershenson, 2019), and effects on occupation choice (Kofoed and McGovney, 2019). While mechanisms are not fully understood, results suggest that role model effects, rather than differences in teacher effectiveness, are driving these disparities (Gershenson et al., 2022).

My work builds upon this prior education literature, and is the first to find outgroup effects in the criminal justice system – reduced arrests and increased employment when paroled individuals are assigned to officers of the same race and sex. As results are found to be driven at the aggregate level, and not by a subset of officers, they fit a hypothesis that results are caused by something akin to role model effects, where societal biases impact the way paroled individuals and parole officers of different races and sexes are able to build trust, relationships, and rapport. From a policy perspective, the results motive a conclusion that these disparities may only be corrected with societal-level changes. However, making parole officers aware of these disparities could potentially help to reduce them.

Section 2 describes the parole system in Pennsylvania and the role of parole officers. Section 3 discusses the data. Section 4 describes the empirical models and estimates for parole officer effects. Section 5 discusses the parole officer survey and the relationship between survey responses and parole officer effects. Section 6 investigates racial and sex outgroup disparities across parolee-officer relationships, and whether they appear to be driven by a subset of "biased" officers. Section 7 concludes.

2. Background – The Parole System in Pennsylvania

Pennsylvania has the sixth highest state parole population, with nearly 50,000 individuals on parole at any given time (Pennsylvania Board of Probation and Parole, 2018). Each paroled

individual falls under the jurisdiction of one of the states' 10 parole district offices, and is assigned to one of 500 community supervision agents (i.e., "officers") in the state. Each officer is assigned a mix of individuals on parole and probation, with an average active caseload of around 60-100 individuals.

A paroled individual is released into the community to serve the remainder of their sentence under community supervision. This supervision is, broadly speaking, defined by three factors. First, the individual is assigned a parole officer to oversee their supervision. Second, the individual is required to meet with their assigned officer on a regular basis.[4] Third, the individual is subject to conditions – such as drug testing, employment requirements, fee payments, alcohol abstention, curfew, program participation requirements, and restrictions on social contacts, that they must abide by at all times while on parole.[5]

Parole officers have wide discretion over the tenor of the parolee-officer relationship and whether and how to sanction paroled individuals for breaking conditions.[6] When structuring supervision, officers must balance the case worker aspects of their job – building rapport with paroled individuals, providing interpersonal support, and connecting them with community re-entry programs and services – with the correctional officer aspects of their job – surveilling the individual, conveying the sanctions associated with future crimes, and determining sanctions after violations. When sanctioning, officers have flexibility over whether to provide a written warning, assign additional parole conditions, require a short-term stay in a non-secure facility, or re-incarcerate. Finally, parole officers have discretion over when and where a paroled individual must meet with them, and when and whether to assign new or remove existing parole conditions.

In Pennsylvania between 2006 and 2020 paroled individuals were assigned to regional parole officer units based on two criteria – the location of the individual's proposed residency and the required special needs of the paroled individual, such as alcohol and other drugs needs, sexual

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[4] In Pennsylvania, individuals are assigned to one of three levels of supervision based on predicted recidivism risk: minimum – one in-person meeting every three months, one collateral contact every three months (a check-in with a close member of the individual's community such as a family member, roommate, or employer); medium – one in-person meeting every month, one collateral contact every three months; or maximum – one in-person meeting every two weeks, one collateral contact every month. For all levels of supervision, every other in-person contact must be at the individual's approved residence. If the individual is in a treatment program, the officer must conduct double the required number of collateral contacts, such that every other contact is with the treatment provider. Collateral contacts may be in-person, over the phone, or over email.

[5] Parole conditions are initially assigned by parole hearing interviewers. After release, parole officers have discretion to impose additional conditions in response to parole violations or remove conditions in response to good behavior.

[6] Officers have wide discretion even among officers in the same parole officer unit under the same supervisor.

offender needs, domestic violence offender needs, and mental health needs. Within each officer unit, paroled individuals were quasi-randomly assigned to parole officers based on officer caseload size at the time of assignment.[7] Specifically, within each parole officer unit paroled individual were assigned to the parole officer in the unit with the smallest caseload at the time of assignment.

From 2006 to 2020 individuals were assigned to parole officers several months before release. At this time the officer would visit and vet the individual's proposed residency (i.e., their "home plan"). If approved, the individual was released into the community to the approved residential address under the supervision of the assigned parole officer who approved the plan. If denied, the individual could propose an alternative residence. If the residence falls within the bounds of a different parole officer unit, the individual was reassigned to an officer in that new unit. This new officer would then visit and vet the new proposed residency, and if approved would oversee the individual's supervision upon release. If at the time of release no proposed residency had been approved, the individual was released to a Community Corrections Center (CCC) (i.e., "halfway house"). Specifically, the individual was released to the CCC with an available room located closest to their most recent proposed residency, and assigned to one of the parole officers associated with that CCC. Officer assignment was quasi-random at CCC's as well – individuals were assigned an officer based on officer caseload size at the time of release and, for certain CCCs, the quasi-random location of their assigned room in the CCC.[8]

3. Data

[7] Based on conversations with Pennsylvania Department of Corrections personnel, during this period officers in some rural units were assigned based on both caseload size and proposed residency location – such that certain officers were more likely to be assigned to certain locations than others. As such, I identify units for which randomization at least partially occurred at the zip code-level, as opposed to the unit-level, as units for which initial officer home plan effects (see Section 4 for details) pass an exogeneity test when constructed at the zip code level but fail an exogeneity test when constructed at the unit-level, with a p-value > .05 (24 units, 36% of the sample). I identify units for which officer assignment was at least partially determined by a more granular geography than zip code as any units for which initial officer home plan effects fail an exogeneity test when constructed at the zip code-level, with a p-value > .05 (13 units, 17% of the sample). For my primary results I consider the randomization group to be at the zip code-level for the first set of units and drop observations in the second set of units, though results are robust to this decision – estimates are appreciable similar when retaining these observations and for alternative randomization group specifications, as shown in Appendix Tables A3 and A7.

[8] Note that paroled individuals are often reassigned to new officers at a later date, due to individuals changing addresses and leaving CCCs, parole officers changing units, or specific parole officer inner-office decisions and requests. In this analysis I focus on initial officer assignment and evaluate intent-to-treat effects. As such, note that the observed officer effect will be a lower bound of the total officer effect if no reassignments took place. Additionally, after one year on parole without incident individuals are eligible for a lower level of "administrative parole" in Pennsylvania. As such I focus on parole officer effects during the first one-year period of parole.

Data on convictions and prison stays, home plans, parole violations, and parole officer-documented employment comes from the Pennsylvania Department of Corrections (DOC). Data on arrests is provided by the Pennsylvania State Police (PSP), and data on tax record-documented employment (from employer unemployment compensation quarterly tax reports) is provided by the Pennsylvania Department of Labor & Industry (L&I).[9]

Table 1 presents summary statistics on individuals paroled to home plans in Pennsylvania, at the time of release. Between March 1st, 2006, and January 1st, 2019, 50,460 individuals were released to parole under home plans.[10] In Pennsylvania, the majority of individuals living in the community under parole supervision are male, and the population is fairly evenly split between non-Hispanic white and Black individuals. Educational attainment and marriage rates are quite low – 35% of individuals have not completed a high school degree, and only 16% are married at the time of release. Convicted crime type is evenly split between violent crimes, drug crimes, and property crimes, and the average remaining sentence length at release to parole is 3.5 years.

Table 1, Panel B presents summary statistics on one-year post-release outcomes. Nearly one-third of individuals paroled to home plans return to state custody within one year of release, via a combination of arrests and technical parole violations (i.e., major violations of parole conditions that trigger re-incarceration). Nearly 40% of individuals receive at least one minor parole violation (i.e., violations of parole conditions that trigger a written warning, new conditions, or placement in a non-secure community facility) and just over half are ever documented as employed during the first year after release.

Figure 1 provides more granular detail on post-release outcomes, over a three-year post-release time horizon. As shown in Panel A, recidivism rates peak in the second quarter after release then slowly fall over time. This reduction is mostly driven by a reduction in major technical parole violations (that trigger re-incarceration), a reduction that occurs mechanically as greater numbers of individuals complete their convicted sentences and move off of parole supervision over time. Panel B shows a similar trend for minor parole violations, which also peak in the second quarter

[9] Each data set is available between 3/06 and 12/19. Specific data set availability ranges are as follows: convictions and incarcerated stays (1/04-6/23); parole hearings (1/04-12/19); home plans (1/05-8/25); parole violations (3/06-6/23); parole officer-documented employment (3/06-3/21); arrests (1/04-5/22); and tax record-documented employment (1/05-12/19).

[10] Appendix Table A1 provides summary statistics on the 50,908 individuals paroled to Community Corrections Centers between 3/1/2006 and 1/1/2019. Overall, a higher percentage of this population was convicted of violent crimes and had higher recidivism risk than individuals paroled to home plans.

after release then slowly fall over time as greater numbers of individuals move off of parole supervision.

Figure 1 Panel C shows notable differences in employment trends when measured using parole officer-documented employment data or tax record-documented employment data. Theoretically, the employment datasets are biased in opposite directions. For parole officer-documented employment, individuals have incentives to overreport employment to comply with the parole condition to be actively working (or searching for work) at all times. For tax record-documented employment, individuals (and employers) have incentives to underreport employment to reduce tax obligations. These biases are likely driving opposing employment trends observed during the first three years after release. Specifically, parole officer-documented employment rates increase each quarter (from 33% to 44%) while tax record-documented employment rates decrease each quarter (from 43% to 32%). These opposing trends fit a hypothesis that, over time, more individuals begin to (1) convince their parole officers that they are employed regardless of actual employment (the incentive to do so grows as parole officers, anecdotally, become more strict about the "actively working" parole requirement over time) and (2) receive their pay under the table (by transitioning from formal employment arranged before release to informal employment). The unbiased employment rate is likely between these numbers – around 40% of paroled individuals are employed each quarter after release.

Figure 1, Panel D presents industry of employment and wage details during the first three years after release, from tax record-documented employment data. Among employed individuals, in the first quarter after release 22% are employed in temporary help positions, 16% in food services, 15% in manufacturing, 10% in construction, and 8% in retail trade. Over the first three years after release the percent employed in temporary help positions decreases (to 11%) while the percent employed in manufacturing and construction slightly increase (to 19% and 13%, respectively). Finally, total quarterly wages are exceptionally low, but increase each quarter (from $3,400 to $5,800) among recently paroled, employed individuals.

Table 2 presents details on the sanctions that are selected for each type of documented parole violation in Pennsylvania. For the 50,460 individuals paroled to home plans between March 2006 and December 2018, 43% of parole violations are for drug test failures, 26% are for broken parole conditions, 23% are for arrests, and 7% are for absconding. Arrest violations almost always

result in reincarceration.[11] Absconding violations result in reincarceration 64% of the time, new conditions 20% of the time, and written warnings 11% of the time. Drug test failures are sanctioned with new conditions 46% of the time, written warnings 24% of the time, non-secure facility stays 16% of the time, and reincarceration 14% of the time. Finally, broken condition violations are mostly sanctioned with written warnings (51%) and new conditions (31%), only occasionally resulting in reincarceration (10% of the time).

4. Parole Officer Effects

Research Design

There are multiple ways to evaluate the extent to which "as-good-as-randomly" assigned individuals – such as police officers, teachers, or supervisors – impact outcomes. My primary specification leverages the literature on randomized decision makers (e.g., Bhuller et al., 2020; Dobbie et al., 2018; Agan et al., 2021; Gross & Baron, 2021; LaForest-Tucker, 2026). This approach creates expected effects for each decision maker – which are often then used as instrumental variables – but are used here to test both the "relevance" of officer effects on outcomes and the "exogeneity" of officer assignments.[12]

As the model includes exogenous covariates (i.e., parole officer units), I construct an officer effect measure for each outcome of interest (arrests, parole violations, and employment) following the effect construction used for Unbiased Jackknife Instrumental Variables Estimation (UJIVE) (Kolesár, 2013). UJIVE effect measures are not biased from the inclusion of covariates, are handled consistently in the first stage, and remain consistent even with a large number of covariates (Chyn et al., (2024)). To create the jackknifed officer effect measure for the parole officer assigned to individual (j), for outcome Y, I first estimate outcome predicted values for every paroled individual other than (j) between March 2006 and December 2018, that take into account both officer fixed effects (O_{it}) and assigned parole officer unit-year fixed effects (U_{it}). These values, $PredY_{it}$, are constructed as the predicted values from an OLS regression of the equation

$$Y_{it} = \gamma_0 + \gamma_1 O_{it} + \gamma_2 U_{it} + e_{it} \quad , \qquad (1)$$

[11] Arrests that do not lead to reincarceration are for low-level summary offenses, i.e., non-traffic citations such as disorderly conduct, harassment, or shoplifting.

[12] Prior literature usually creates expected decision maker effects as instrumental variables to proxy for decisions of interest, such as how the decisions of whether to incarcerate (Bhuller et al., 2020) and whether to parole (LaForest-Tucker, 2026) impact future outcomes. Here, I create expected officer effects directly for each future outcome to evaluate the extent to which relative officer effects for future outcomes vary.

using all observations $i \neq j$, where e_{it} is a stochastic error term.[13] Next, I partial out U_{it} from these predicted values, to create an unbiased measure of each officer's expected outcome effect. Specifically, the officer effect for paroled individual (j) – V_{jt} – is constructed as the predicted residuals from an OLS regression of the equation

$$PredY_{it} = \gamma_0 + \gamma_1 U_{it} + e_{it} \quad , \qquad (2)$$

using all observations $i \neq j$. Once constructed, the jackknifed officer effect measure for each outcome Y can be used to test (1) the magnitude of each officer's expected effect relative to other officers (i.e., relevance), and (2) the extent to which officer effects appear independent from observable characteristics about the individual (X_{it}) (i.e., exogeneity).[14]

In addition to varying how they provide supervision, note that officers may vary in their propensity to approve proposed residences. Even if officers are quasi-randomly assigned prior to release, if home plan approval rates vary then officer assignments *at time of release* may no longer be random. Appendix Table 1 tests the relevance and exogeneity of initially assigned agent home plan approval effects. While initial assignments appear to be random (a formal joint f-test for independence between jackknifed officer home plan approval effects and other observable characteristics about the individual (X_{it}) produces a p-value of 0.56), officers appear to approve home plans at different rates (the f-value for the relationship between the leniency measure and home plan approval, controlling for all other observable characteristics about the individual, is 411). Note, however, that officer initial approval rates have at most a modest impact on whether an individual is released to a home plan of any kind relative to being released to a CCC, as individuals whose home plans are initially denied usually have time to submit an alternative address for approval prior to release (the f-value for the relationship between initial officer home plan leniency and eventual release to a CCC, controlling for all other observable characteristics about the individual, is 57).

Still, if officer home plan leniency is correlated with officer expected outcome effects, the subset of individuals released to home plans may be unbalanced across officers. Due to this

[13] The primary specification drops (1) officers assigned less than 20 total paroled individuals or 20 total initial home plan decisions and (2) officer unit-year combinations where only one officer was ever assigned. Results are appreciably similar for alternative specifications, as shown in Appendix Tables A3 and A7.

[14] X_{it} is comprised of the following covariates: male, black, Hispanic, married, divorced / separated, high school graduate, age at release, violent crime conviction, drug crime conviction, public order crime conviction, minimum sentence length, whether paroled at minimum sentence hearing, remaining sentence length at release, medium recidivism risk, and high recidivism risk.

concern I estimate two primary specifications. The first assumes no such bias and considers randomization groups U_{it} to be comprised of only parole officer unit and year. The second directly incorporates officer home plan leniency into randomization groups U_{it}.[15] This specification creates exogenous "outcome effect" measures that compare individuals in the same parole officer unit and year, who were released to officers with similar home plan approval leniencies. This specification recovers causal, unbiased estimates of relative officer effects on post-release outcomes, within these randomization groups (Humphries et al., 2025; LaForest-Tucker, 2026). Note, however, that controlling for officer home plan leniency in randomization group fixed effects U_{it} (which are now substantially more granular) reduces the available sample size, as individuals in randomization groups U_{it} without officer variation are dropped.

An alternative approach – to evaluate the extent to which parole officer assignment impacts post-release outcomes – is to compare the extent to which individual officer fixed effects impact outcome estimates, relative to other observable characteristics about the paroled individual (Weisburst, 2024). Specifically, let Y_{it} be an outcome for individual i after release at time t, let X_{it} be a set of observable personal characteristics about the paroled individual and level of parole supervision, let U_{it} be the paroled individual's parole officer unit and year released, and let ε_{it} be a stochastic error term. The impact of observable personal characteristics on post-release outcomes, without taking into account individual parole officer effects, is given by vectors β_1 and β_2 in Equation 3:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \varepsilon_{it} \quad . \tag{3}$$

Next, let $Agent_i^j$ equal one if individual i is assigned to parole officer j at the time of release, and zero otherwise. The impact of observable personal outcomes when taking into account parole officer effects is given by vectors β_1, β_2, and β_{3k} for each parole officer j in Equation 4:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \sum_k \left(\beta_{3k} Officer_i^k\right) + \varepsilon_{it} \quad . \tag{4}$$

Additionally, note that the adjusted R^2 associated with an OLS regression of Equation 3 (AR_1^2) conveys the percent of variation in outcomes that can be ascribed to observable characteristics not including parole officer. Note, additionally, that the adjusted R^2 associated with an OLS regression of Equation 4 (AR_2^2) conveys the percent of variation in outcomes that can be ascribed

[15] The primary analysis splits officer home plan leniency into three bins - lenient, neutral, and strict - in randomization groups U_{it}. As shown in Appendix Table A3, results are appreciable similar for alternative leniency group specifications.

to observable characteristics including parole officer. As such, $AR_2^2/AR_1^2 - 1$ represents the percent of total observed variation in outcomes attributable to assigned parole officers.

Results

Results for relevance and exogeneity tests, using jackknifed expected effect measures, are presented in Table 3. Panel B presents relevance and exogeneity tests for minor parole violation effect measures, for specifications excluding and including jackknifed home plan leniency effects in randomization groups U_{it}. For each specification, the measures pass exogeneity tests – formal joint f-tests for independence between the measures (V_{jt}) and other observable characteristics about the individual (X_{it}) produce p-values of 0.32 and 0.99, respectively. For both specifications, the parole violation effect measures are also quite relevant – first stage f-values for the relationship between the expected parole violation effect measure and whether a parole violation occurred, controlling for all other observable characteristics about the individual (X_{it}) and assigned randomization group (U_{it}), are 447 and 174, respectively.

Panel A presents the results from relevance and exogeneity tests for every other outcome of interest. For effect measures created without including officer home plan leniency in randomization groups U_{it}, exogeneity results vary across outcomes with p-values between .00 and .86, with an average p-value of 0.16. For effect measures created including officer home plan leniency in randomization groups U_{it}, measures appear exogenous and unrelated to all observation characteristics about the individual for each outcome, with p-values between .03 and .99 with an average p-value of .42. The relevance of officer effect measures varies significantly across outcomes. Officers appear to have little effect on arrests (F-stats of 7 and 4), a slight effect on parole violation reincarceration (F-stats of 51 and 18), and a substantial effect on each type of minor parole violation sanction (F-stats ranging from 83-446). Finally, officers have a substantial effect on officer-documented employment (F-stats of 822 and 367) but almost no effect on tax-record-documented employment (F-stats of 11 and 4).[16]

These relevance results are shown graphically in Figure 2. Each graph presents a histogram of officer expected effects for an outcome, along with a local linear regression (LLR) of an

[16] Results are appreciably similar for a wide variety of alternative specifications, as shown in Appendix Table A3. Results for the effects of parole officers for individuals released to community corrections centers are presented in Appendix Table A4. Officer assignment appears exogenous, but different officers appear to have little effect on outcomes for individuals released to CCCs.

individual's outcome on assigned parole officer expected effect for that outcome. While all histograms appear normally distributed, the local linear regressions for arrest and tax record-documented employment are nearly flat, conveying that officers have little impact on these outcomes.[17] Conversely, the local linear regressions for minor parole violations and officer-documented employment show very strong relationships. For each, the LLRs are close to monotonically increasing with relatively high slopes. These graphs convey that moving from the officer with the lowest expected parole violation effect to the highest expected parole violation effect causes a 25 percentage point increase (from 25% to 50%) in the likelihood of incurring a minor parole violation in the first year after release. Similarly, moving from the officer with the lowest expected officer-documented employment effect to the highest expected officer-documented employment effect leads to a 25 percentage point increase (from 37% to 62%) in the likelihood of ever being documented as employed during the first year after release.[18]

Results for "officer fixed effects" regressions reach the same conclusion, as shown in Table 4. Columns 1 and 2 present the Adjusted R^2's for regressions of different outcome variables, over the first year after release to parole, excluding and including officer fixed effects, respectively. As shown in Column 1, all known information about an individual on parole, other than assigned parole officer, can only explain 6-12% of post-release outcomes. Controlling for one's quasi-randomly assigned parole officer increases R^2 by somewhere between zero and four percentage points, depending on the outcome, as shown in Column 3. Column 4 presents the percent of adjusted R^2 attributable to officer fixed effects. Overall, officers appear to have little relative effect on post-release recidivism, only explaining 3% and 9% of the explainable variation in arrest and parole violation reincarceration outcomes, respectively. However, officers appear to have a large relative effect on documented minor parole violations (that result in written warnings, new restrictions, or placement in a non-secure facility, but do not result in reincarceration), explaining 24% of the explainable variation in minor parole violations in the first year after release. While officers have a large relative effect on officer-documented employment (explaining 36% of the

[17] Specifically, flat LLR's convey that officers' jackknifed expected effect measures are not correlated with actual outcomes.

[18] Graphs depicting the relevance of expected officer effects on specific violation types (broke condition, failed drug test, and absconding), as opposed to specific sanction types, are appreciably similar and presented in Appendix Figure A1.

explainable variation) they have little effect on tax record-documented employment (explaining only 2% of the explainable variation).

Overall, it appears that individual parole officers do not have large effects on recidivism, either by decreasing future arrests or major parole violations that cause re-incarceration. However, individual officers do appear to have large effects on minor parole violations that lead to written warnings, new conditions, and stays in non-secure facilities. This effect could be caused by several mechanisms. First, certain officers may build relationships with paroled individuals that cause them to change their post-release behavior, impacting their propensity to break minor conditions of parole. Alternatively, officers may have no effect on paroled individual behavior. Instead, officer effects on minor violations may be driven by differences in officer reporting and sanctioning decisions. For example, certain officers may be more likely to write up and sanction individuals for violations, even if they have no direct impact on the paroled individual behavior that causes those violations. These results could be driven by officer effects on individual behavior, officer differences in reporting and sanctioning decisions, or a combination of both.

The analysis also concluded that individual officers have a large effect on officer-documented employment but little effect on tax record-documented employment. This result supports a conclusion that officer effects are driven by the second mechanism discussed above – differences in officer reporting and sanctioning decisions – as opposed to officer effects on paroled individual behavior. As officers, individually, have little effect on tax record-documented employment, it is likely that they have little effect on paroled individuals' propensities to find and retain employment after release to the community. Instead, the results support the hypothesis that officer effects on officer-documented employment are driven by officer differences in how they follow up, vet, and document reported employment. This result also supports a hypothesis that officer effects on minor parole violations are driven by officer differences in how they document and sanction individuals for parole violations, and not by effects on individual behavior.[19]

Reductions in minor parole violations are still beneficial for paroled individuals even if they are caused by sanctioning differences as opposed to impacts on paroled individual behavior. Specifically, each marginal sanction that adds one or more new parole conditions increases the

[19] The correlation between officer expected effects on minor parole violations and officer-documented employment is .25. This correlation implies that assignment to an officer that decreases documented minor violations also decreases officer-documented employment, at least to an extent.

burden of parole supervision without improving future recidivism or employment outcomes (LaForest-Tucker, 2026). As such, finding ways to reduce minor violations – even if only through changes to officer sanctioning and reporting decisions – is beneficial for paroled individuals.

5. Interpersonal Support, Structural Support, and Deterrence

I next investigate the relationship between parole officer effects and officers' self-reported approaches to parole supervision, as documented by a survey of parole officers. In Pennsylvania, the Department of Corrections surveyed all parole officers in three consecutive years – 2015, 2016, and 2017. The survey included 37 questions and statements about parole officers' job perspectives, such as "I build rapport with the parolees on my caseload", "I want to help parolees", "Reentry is part of supervision", and "I consider myself more of a case worker than a police officer", scored on a scale of 1 (strongly disagree) to 5 (strongly agree).[20] While parole officers were not required to complete the survey, response rates were fairly high – on average 59% of parole officers responded to the survey each year.

The 37 questions fall into 11 categories. The first five cover officers' perspectives on the interpersonal relationships between themselves and paroled individuals. "*Relationship Important*" questions refer to whether officers prioritize having a good relationship with their paroled individuals, good rapport, and whether they think that their paroled individuals feel comfortable talking to them about issues. *"Care about Parolees"* questions refer to the extent to which they care about paroled individuals and want to help them, and *"Treat Nonjudgmentally"* questions refer to whether they use a nonjudgemental approach in their interactions, are optimistic, and praise paroled individuals for success. *"Support after Mistake"* questions refer to whether, after a violation or other failure, they work with the individual to find out what caused the failure and talk with them before taking serious action. Finally, "*Learn about Parolees*" questions refer to the extent they try to learn about paroled individuals and think about things from their points of view.

The next four categories cover officers' perspectives about providing structural support. *"Match to Suitable Interventions"* questions refer to whether they can identify individuals' needs, and match programing and interventions to their risk levels. "*Encourage Personal Improvement*" questions refer to whether officers try to help individuals learn pro-social skills, pursue personal improvement, and provide mentoring. "*Believe Parolees can Change*" questions refer to whether

[20] See Appendix B for the full set of survey questions.

they believe paroled individuals are capable of change and learning pro-social behavior, and *"Contact Support System"* is a single question about how frequently officers contact individuals' support systems.

Finally, "*Case Worker not Police Officer*" questions refer to the extent officers prioritize social work (i.e., interpersonal and structural support) over law enforcement (i.e., deterrence). "*Favorable Opinion of Parole Policies*" questions refer to whether officers think current Pennsylvania parole policies help the reentry process and whether current workloads are manageable, and "*Responded*" is an indicator variable for whether each officer ever responded to the survey.

Let S_i represent the vector of survey responses for the parole officer assigned to paroled individual i. The relationship between parole officer perspectives S_i and individual i's outcome Y_{it} is represented by β_s in Equation 6,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_3 H_{it} + \beta_s S_i + \varepsilon_{it} \quad , \qquad (5)$$

where U_{it} is the paroled individual's assigned parole officer unit-year, H_{it} is the jackknifed home plan leniency of the assigned parole officer, and X_{it} is a vector of other observable characteristics about the individual.[21]

Summary statistics for aggregate survey categories are presented in Table 5, Column 1.[22] On average, officers that responded to the survey care about building interpersonal relationships with paroled individuals – responding with an average of 4.0 out of 5 across interpersonal relationship-related questions. Officers also prioritize providing skill-building and structural support to paroled individuals, responding with an average of 3.7 across questions about providing structural support. Next, on average officers tend to focus more on support than deterrence, with an average of 3.7 on questions related to whether their supervision is focused on reentry rather than law enforcement. Finally, on average officers that responded to the survey had a moderate opinion of parole policies, responding with an average of 2.5 across questions about whether workload sizes are manageable and parole policies are supportive.

The impact of parole officer job perspectives on outcomes, as captured by survey responses, is also presented in Table 5. First, I find little effect of survey responses on arrests,

[21] As the survey was only conducted from 2015 and 2017 I do not allow survey responses to vary over time. For parole officers who filled out the survey in multiple years the first completed survey is used.
[22] Summary statistics and results for each individual survey question are presented in Appendix Table A5.

supporting the conclusion that assigned officers have little relative effect on arrests in Pennsylvania. Regarding parole violations, I find that officers that focus on interpersonal relationships reduce parole violations – those that care about building good relationships reduce minor violations (by 2.0 percentage points for a one standard deviation increase in response), and those that focus on being nonjudgemental and supporting paroled individuals after mistakes reduce both minor (by 1.1 and 1.4 percentage points for a one standard deviation increase in response) and major (by 1.3 and 1.1 percentage points) violations.

Officers that focus on providing skill-building and structural support appear to have little relative effect on paroled individuals, though results suggest that assignment to these officers leads to a slight increase in minor parole violations. However, officers that report having frequent contact with paroled individuals' support systems appear to slightly increase both officer-documented and tax record-documented employment (by 0.8 and 1.4 percentage points for a one standard deviation increase in response, respectively). Officers that focus on support rather than deterrence have no relative effect on arrests, but appear to slightly increase minor parole violations (1.4 percentage points) and tax record-documented employment (1.0 percentage points). Next, individuals assigned to officers who believe current Pennsylvania parole policies are reasonable have higher officer-documented and tax record-documented employment rates (by 3.5 and 1.2 percentage points for a one standard deviation increase in response, respectively). Finally, individuals assigned to officers who responded to the optional survey have 1.2 and 1.6 percentage point lower major and minor parole violation rates.

These results show that individuals assigned to parole officers that focus on building interpersonal connections reduce minor parole violations. While the effect is likely driven by differences in officer sanctioning and reporting behavior, as opposed to changes in paroled individual behavior, additional minor violations are still detrimental for individuals as they increase the burden of parole supervision without positively impacting future outcomes (LaForest-Tucker, 2026). As such, the results suggest that officers that focus on interpersonal support improve post-release outcomes for paroled individuals relative to officers that focus on structural support or deterrence.

Note that it is not clear whether the effect of focusing on interpersonal support is caused by an officer's decision to focus on interpersonal support, or by other unobserved characteristics of officers that choose to focus on interpersonal support. For example, reductions in minor parole

violations could be caused by officers choosing to focus on interpersonal support. If so, these results motivate a policy recommendation centered on officer training – more officers should be trained to prioritize interpersonal support. Alternatively, these effects could be driven by unobserved characteristics of certain parole officers, which they bring to the job with them when they are hired, who prefer to focus on interpersonal support. If so, these results motivate a policy recommendation centered on officer hiring – more officers should be hired who are interested in prioritizing interpersonal support.

6. Outgroup Disparities and Bias

Next, I investigate the relationship between parole officer demographic characteristics, paroled individual demographic characteristics, and outcome effects. First, the relationship between parole officer observable characteristics, P_{it}, and assigned paroled individual outcomes is represented by β_p in Equation 7,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_3 H_{it} + \beta_p P_{it} + \varepsilon_{it} \quad , \qquad (6)$$

where P_{it} is comprised of parole officer race, sex, and years of service variables.

The relationship between parole officer demographic characteristics and assigned paroled individual outcomes is presented in Table 6. Being assigned a male or black officer has no effect on recidivism or tax-record-documented employment, but each lead to a decrease in minor parole violations and parole-officer documented employment, relative to being assigned a female or white officer. Individuals assigned to officers with additional years of service are less likely to receive major or minor parole violations, and more likely to be documented by their officer as employed, during their first year post-release. Specifically, individuals assigned officers with one additional year of service are 1.5% less likely to have a major parole violation that results in reimprisonment, and 0.5% less likely to have a minor parole violation (that results in a written warning, additional parole conditions, or stay in a non-secure facility). These results fit a hypothesis that female officers, white officers, and newer officers may be assigning more sanctions in order to build up authority and agency with their assigned paroled individuals.

I next explore the impact of differences in relative parolee-officer race and sex. Let Q_{it} represent a vector of the relative race and sex quadrant indicators of the parolee-officer relationship for paroled individual i ("white-white" / "white-nonwhite" / "nonwhite-nonwhite" / "nonwhite-

white"; "male-male" / "male-female" / "female-female" / "female-male"). The effect of relative parolee-officer race and sex is represented as β_q in Equation 8,

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 U_{it} + \beta_3 H_{it} + \beta_q Q_{it} + \varepsilon_{it} \quad . \tag{7}$$

Results on the average effect of relative parolee-officer race, and relative parolee-officer sex, are presented in Table 7.[23] Panel A presents percentage point effects and Panel B presents raw estimates. At the aggregate level, being assigned a parole officer of the opposite race (non-Hispanic white vs. Black or Hispanic) has a significant impact on recidivism. Specifically, white individuals assigned to non-white officers have a 7% higher recidivism rate, on average, than white individuals assigned to white officers (after taking into account parole officer unit, parole officer home plan leniency, and observable demographic characteristics about the paroled individual). Similarly, non-white individuals assigned to white officers have an 11% higher recidivism rate than non-white individuals assigned to non-white officers, leading to a net 18% increase in recidivism from being assigned to an officer of the opposite race. While not statistically significant, results also suggest that being assigned an officer of the opposite race leads to increased minor parole violations and decreased officer-documented and tax record-documented employment.

The analysis also finds a significant effect of relative parolee-officer sex. Specifically, female officers increase officer-documented employment for both male and female individuals, but increase it substantially more for female individuals (9% and 19%, respectively). Female officers have little effect on tax record-documented employment for men, but increase it by 10% for women. Thus, across both measures, the analysis concludes that assignment to an officer of the opposite sex reduces employment by 10%. While not statistically significant, results also suggest that being assigned an officer of the opposite sex leads to increased arrests and parole violations.

These results find aggregate-level outgroup effects in parolee-officer relationships. Note, however, that these effects could be driven by a variety of factors. They could be driven by parole officer racial and gender bias. For example, some non-white officers could be treating white individuals more harshly than they should be, or black individuals more leniently than they should be. Alternatively, some white officers could be treating white individuals more leniently, or black

[23] Results are appreciably similar for alternative specifications and samples, as shown in Appendix Table A7.

individuals more harshly, than they should be. Any of these four hypothesis are consistent with observed aggregate outgroup disparities. Alternatively, the disparities could be driven by how individuals themselves *respond* to being assigned parole officers of a different race or sex. That is, individuals might respond differently to officers of different races, even if the officers themselves are treating all assigned individuals equally. Finally, these results could be caused by a mix of these factors, driven by aggregate-level structural or societal biases, that cause perceived differences in race and sex to impact the strength and quality of the supervisory relationship parole officers and paroled individuals are able to, jointly, construct. It is unclear from these aggregate estimates whether the outgroup disparities are driven by bias among certain parole officers, bias in paroled individual behavior, or combined systemic bias that impacts the relationships that individuals and parole officers are able to construct.

To investigate these potential mechanisms, I explore the extent to which aggregate-level racial and gender outgroup disparities are driven by a subset of officers.[24] To do so, I construct officer-level outgroup expected effect measures, for each outcome, for parolee-officer relative race and parolee-officer relative sex. To construct a racial outgroup effect measure for each outcome, I first construct officer expected effect measures following Equations 1-2, that allow officer effects to differ across parolee race. I then define the "white/non-white relative effect" for each officer as the difference in that officer's average jackknifed effect for white parolees and the officer's average jackknifed effect for non-white parolees. I construct officer-level "male/female relative effect" outgroup effect measures analogously.

Distributions of these "white/nonwhite relative effect" and "male/female relative effect" measures on arrests, minor parole violations, and tax-documented employment are displayed in Figure 3.[25] Figure 3, Panel A presents the distribution of the "white /non-white relative effect" for

[24] I separately test whether there is evidence of monotonicity in officer effects across parolee race and sex subgroups. Following (Bhuller et al., 2020), I investigate (1) within-subsample monotonicity by testing relevance for jackknifed effects constructed using the full sample, in regressions restricted to each race and sex subsample, and (2) across-sample monotonicity by testing relevance for jackknifed effects constructed omitting each subsample, in regressions restricted to the omitted subsample. Results are presented in Appendix Table A7. Overall, there is no evidence of a break in monotonicity within or across subsamples, for officer effects on minor violations and documented employment. Generally, these results show that officers who increase parole violations and officer-documented employment for white individuals and male individuals have at least a somewhat similar effect non-white individuals and female individuals.

[25] Results are appreciably similar for alternative "relative effect" specifications, such as observations presented at the paroled-individual-stay level (i.e., multiple observations per parole officer), and effects constructed using inverse probability weights, as shown in Appendix Figure A2.

each parole officer (top histogram), each white parole officer (middle histogram), each non-white parole officer (bottom histogram), and a quantile-quantile plot (Wilk and Gnanadesikan, 1968) comparing the effect distributions for white and nonwhite parole officers. First, these figures show no visual evidence of outlier officers with very large "white/non-white relative effects" among either white or nonwhite officers – the distribution of these effects for both white and non-white officers appear similarly distributed to the distribution for all officers. However, the means of the effect distributions for white and nonwhite officers differ slightly. Specifically, the white officer distribution has a slightly lower mean than the non-white officer distribution for arrest and minor violation outcomes, conveying that, on average, white officers have slightly more negative arrest and parole violation effects for white parolees than non-white parolees, relative to non-white officers. The same is true, in the other direction, for tax record-documented employment – the white officer distribution has a slightly higher mean than the non-white officer distribution for employment effects, conveying that, on average, white officers have slightly more positive employment effects for white parolees than non-white parolees, relative to non-white officers. These visual conclusions are confirmed by the quantile-quantile plots, which depict non-white officers having lower relative arrest effects than white officers at every quantile of their respective distributions. Overall, the outgroup effects do not appear to be driven by a subset of outlier officers, but instead by a slight shift in the mean of the entire distribution between white and nonwhite officers.

Figure 3, Panel B, presents distributions for sex outgroup effects. Results are quite similar to those found in Panel A. Figures show no evidence of outlier officers, with the main difference in the distributions of "male/female relative effects" for male and female officers being a slight shift in the mean of the entire distribution. Specifically, the mean of the distribution for male officers is lower for arrests and parole violations (i.e., reduced arrests and violations for male parolees relative to female parolees) and higher for employment (i.e., increased employment for male parolees relative to female parolees) compared to the mean of the distribution for female officers. Quantile-quantile plots confirm this conclusion – male officers have higher relative employment effects than female officers at every quantile of their respective distributions.

Overall, these results are consistent with the hypothesis that observed racial and sex outgroup disparities in parolee-officer pairings are not caused by a subset of biased parole officers (i.e., "a few bad apples"). The disparities observed in the aggregate data appear to be fairly evenly

distributed across officers, consistent with a hypothesis that these effects are driven by larger, systemic or structural biases that impact the way paroled individuals and parole officers of different races and sexes are able to build trust and relationships. Overall, they fit a hypothesis that something akin to role model effects, which potentially drive student-teacher outgroup disparities (Gershenson et al., 2022), occur within parolee-officer relationships. They motive a conclusion that these disparities may not be easily fixed by changes in parole policy, and may instead only be corrected by changes at a societal level. However, increasing parole officer awareness of these disparities could potentially help to reduce them.

7. Discussion

This work shows that individual parole officers do not appear to have a notable relative effect on recidivism – whether through future arrests or parole violations that lead the reincarceration. However, individual officers do appear to have an effect on minor parole violations – that lead to written warnings, additional parole conditions, and stays in non-secure facilities. These effects are also important, as minor parole violations often lead to additional conditions which a paroled individual must abide by, that increase the burden of supervision. Based on parole officer survey responses, officers that prioritize interpersonal supports reduce violations, providing suggestive evidence that parolee-officer relationships that focus on interpersonal supports can be beneficial for paroled individuals. However, officers that prioritize structural supports do not appear to have a notable effect on outcomes, providing suggestive evidence that existing support services are not effective in facilitating reentry.

This work also finds evidence of racial and gender outgroup disparities in parolee-officer relationships. Specifically, being assigned a parole officer of a different race or gender leads to higher rates of recidivism, higher rates of minor parole violation, and lower employment. However, there is no evidence these disparities are driven by individual officers. Instead, the evidence supports a hypothesis that these effects are driven by structural or societal biases that negatively impact the ability of a paroled individual and parole officer to build a strong and supportive relationship. As such, the results motive a conclusion that these biases must be tackled at a more aggregate societal scale. However, parole officers should be aware of these disparities so that they can actively work to reduce them when building parolee-officer relationships.

Overall, these results support a conclusion that the parolee-officer relationship has a nuanced effect on the outcomes of paroled individuals. For example, employing more officer that prioritize building interpersonal connections with paroled individuals could reduce minor parole violations. However, the conclusion from this work reinforce the results of other work on the effects of community corrections and reentry programs – that these programs and supervision policies appear to have little impact on overall returns to custody. As such, more research is needed to test alternative programs and policies that could appreciably improve the reentry outcomes of paroled individuals.

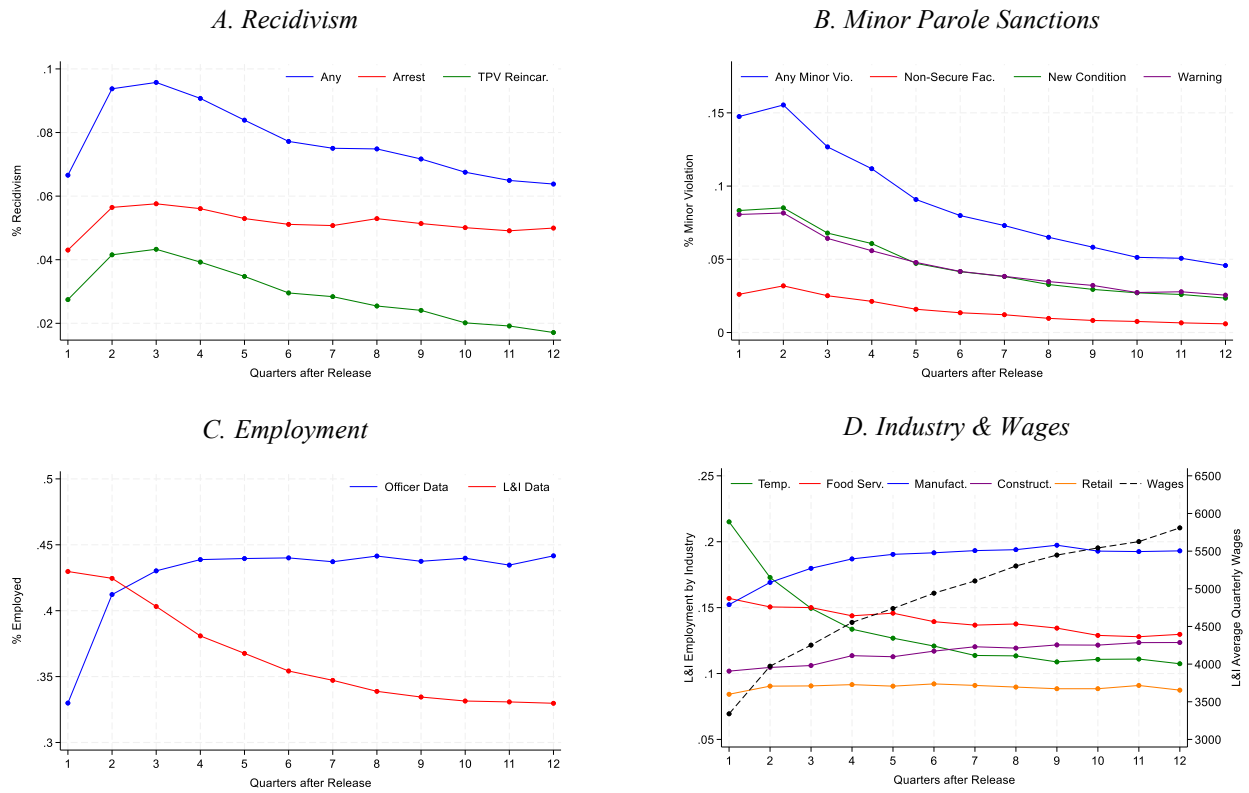
Tables & Figures

Table 1 – Paroled Population

	Mean
A. Demographic Characteristics	
Male	88%
Black	38%
Hispanic	8%
Married	16%
Divorced / Separated	11%
No High School Degree	35%
Age at Release	35.0
Violent Crime Conviction	25%
Drug Crime Conviction	34%
Property Crime Conviction	20%
Minimum Sentence Length (Years)	2.5
Paroled at Minimum Sentence Hearing	67%
Remaining Sentence at Release (Years)	3.5
Medium Recidivism Risk Level	42%
High Recidivism Risk Level	29%
B. Outcomes – One Year After Release	
Recidivism *(Arrested or Reincarcerated)*	29%
Arrest	20%
Parole Violation Reincarceration	10%
Minor Parole Violation	39%
Ever Employed (Parole Data)	55%
Ever Employed (L&I Data)	61%

Notes: This table presents summary statistics for the 50,460 home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2019.

Figure 1 – Outcomes by Quarter



Notes: Figure depicts three years of average post-release outcomes, by quarter, for all home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2017. Graph A presents arrest and non-arrest technical parole violation reincarceration outcomes. Graph B presents minor parole violation outcomes, broken out by minor violation type – non-secure facility stays, new conditions, and written warnings. Graph C compares quarterly employment rates from parole officer-documented employment data and PA Department of Labor and Industry-documented employment data. Graph D presents industry-specific employment rates and average wages - among the subset of employed individuals - from PA Department of Labor and Industry-documented employment data. N = 39,455 for all outcomes except parole officer-documented employment. Note that parole officer-documented employment sample sizes decrease from Q1-Q12 (N = 36,119 in Q4, 27,493 in Q8, and 19,836 in Q12) as some individuals reach their max sentence date and are no longer on parole.

Table 2 – Parole Violations & Sanctions

	Sanction Type				
	Written Warning	New Conditions	Non-Secure Facility Stay	Reincarc.	**Total**
Violation Type					
Broke Condition	51%	31%	4%	10%	37,983
Failed Drug Test	24%	46%	16%	14%	64,343
Absconded	11%	20%	4%	64%	9,429
Arrested	3%	14%	2%	81%	34,329
Assaultive Behavior	4%	24%	4%	67%	1,406
Weapon Possession	14%	21%	6%	59%	659
Total	36,833	48,241	12,705	47,907	

Notes: Percentages aggregate from left to right. This table presents summary statistics for all parole violations that occur before 6/1/2023 for the 50,460 home plan releases to parole in Pennsylvania between 3/1/2006 and 1/1/2019. N = 148,171.

Table 3A – Relevance & Exogeneity Summary for Officer Outcome Effect Measures

	Primary Outcome Propensity Measures				**Propensity Measures Conditional on Home Plan Leniency**			
	Relevance Test		***Exogeneity Test***		***Relevance Test***		***Exogeneity Test***	
	F-Stat.	*P-Val.*	*F-Stat.*	*P-Val.*	*F-Stat.*	*P-Val.*	*F-Stat.*	*P-Val.*
Recidivism (*arrested or reincar.*)	10.7	*(.00)*	2.5	*(.00)*	2.0	*(.16)*	1.3	*(.18)*
Arrest	8.0	*(.00)*	1.9	*(.01)*	4.5	*(.03)*	1.1	*(.32)*
PV Reincarceration	55.0	*(.00)*	1.5	*(.07)*	20.0	*(.00)*	1.4	*(.12)*
Minor Parole Violation	434.6	*(.00)*	1.2	*(.21)*	169.6	*(.00)*	0.4	*(.99)*
Sanction Type								
Written Warning	435.8	*(.00)*	0.7	*(.82)*	265.0	*(.00)*	0.4	*(.98)*
New Condition	304.2	*(.00)*	1.6	*(.04)*	108.9	*(.00)*	0.8	*(.68)*
Non-Secure Facility Stay	82.2	*(.00)*	1.6	*(.05)*	76.8	*(.00)*	1.1	*(.30)*
Ever Employed (Parole Data)	810.9	*(.00)*	2.3	*(.00)*	372.0	*(.00)*	1.2	*(.30)*
Ever Employed (L&I Data)	15.7	*(.00)*	2.4	*(.00)*	5.9	*(.01)*	1.7	*(.03)*

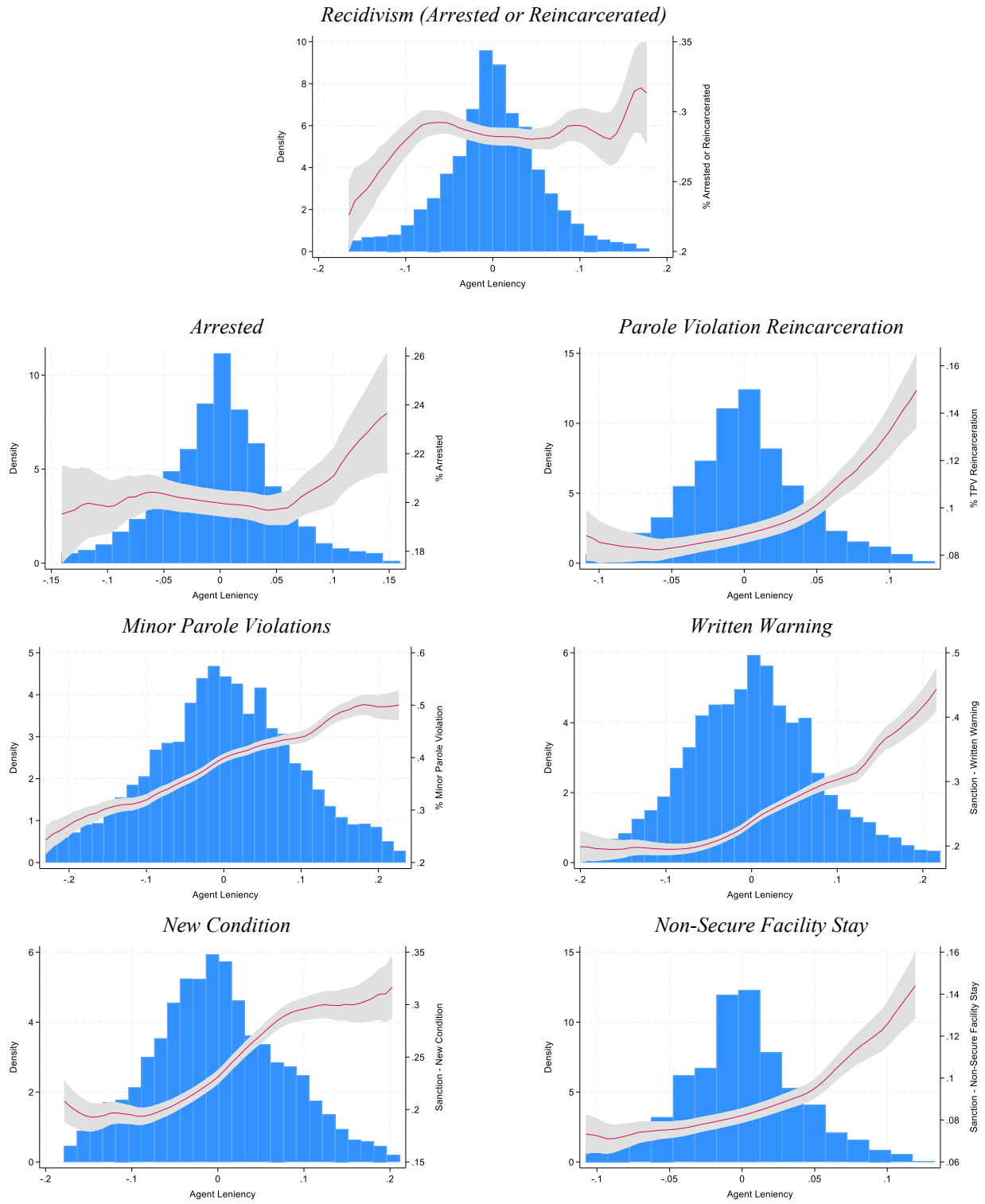
Notes: Columns 1 & 3 present f-test values for whether β(propensity) = 0, from regressions of whether each outcome occurred during the first-year post-release on each officer outcome propensity measure, controlling for observable characteristics about the individual. Columns 2 & 4 present joint f-test p-values from regressions of officer outcome propensity measures on observable characteristics about the individual. When constructing officer outcome propensities, Columns 3 & 4 include three agent home plan leniency types (lenient, neutral, and strict) in randomization groups U_{it}, and drop randomization groups for which there is no longer sufficient officer variation. N = 29,265 for Columns 1 and 2 and N = 22,214 for Columns 3 and 4.

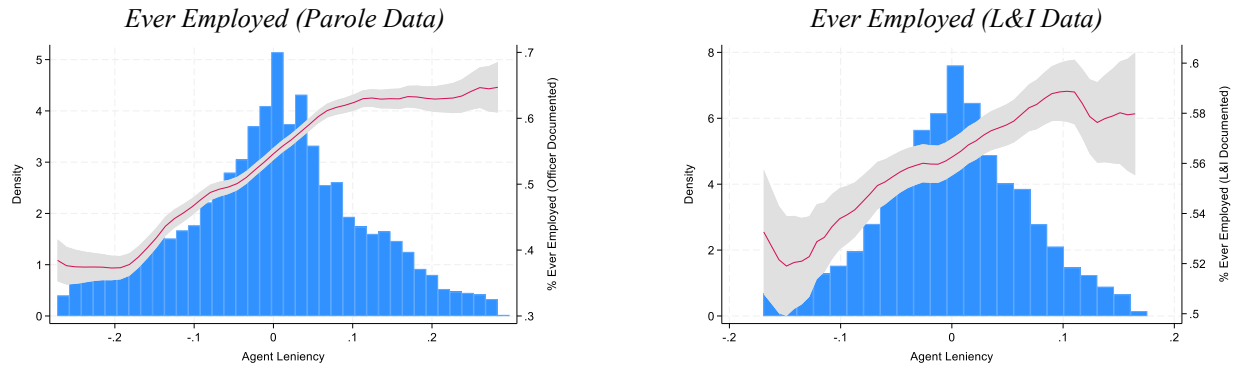
Table 3B – Relevance & Exogeneity Details for Officer Minor Parole Violation Effect Measure

	Primary Outcome Propensity Measure				**Propensity Measure Conditional on Home Plan Leniency**			
	Relevance Test		***Exogeneity Test***		***Relevance Test***		***Exogeneity Test***	
	Est.	***SE***	***Est.***	***SE***	***Est.***	***SE***	***Est.***	***SE***
Dependent Variable	***Minor Violation Occurred***		***Minor Violation Propensity***		***Minor Violation Occurred***		***Minor Violation Propensity***	
Minor Violation Propensity	.5130 ***	*.0246*	-	-	.3705 ***	*.0285*	-	-
Male	.0162 *	*.0092*	-.0017	*.0023*	.0145	*.0107*	.0010	*.0026*
Black	.0291 ***	*.0080*	-.0029	*.0020*	.0301 ***	*.0093*	-.0007	*.0023*
Hispanic	.0090	*.0120*	-.0006	*.0029*	.0122	*.0138*	.0035	*.0034*
Married	-.0331 ***	*.0081*	-.0030	*.0020*	-.0340 ***	*.0093*	.0001	*.0023*
Divorced / Separated	-.0005	*.0101*	-.0040	*.0025*	-.0113	*.0117*	.0016	*.0029*
High School Graduate	-.0165 ***	*.0059*	.0011	*.0015*	-.0171 **	*.0068*	.0000	*.0017*
Age at Release	-.0040 ***	*.0003*	.0000	*.0001*	-.0043 ***	*.0003*	.0000	*.0001*
Violent Crime Conviction	-.0808 ***	*.0093*	.0028	*.0023*	-.0856 ***	*.0107*	-.0003	*.0026*
Drug Crime Conviction	-.0585 ***	*.0084*	.0057 ***	*.0021*	-.0562 ***	*.0097*	.0009	*.0024*
Minimum Sentence Length (Years)	-.0115 ***	*.0020*	.0003	*.0005*	-.0125 ***	*.0023*	-.0001	*.0006*
Paroled at Minimum Hearing	-.0115 *	*.0064*	-.0002	*.0016*	-.0136 *	*.0073*	.0002	*.0018*
Remaining Sentence (Years)	-.0029 **	*.0014*	-.0003	*.0003*	-.0020	*.0016*	.0001	*.0004*
Medium Recidivism Risk Level	.1153 ***	*.0072*	.0004	*.0018*	.1121 ***	*.0083*	.0022	*.0020*
High Recidivism Risk Level	.1955 ***	*.0079*	.0006	*.0019*	.1926 **	*.0091*	.0021	*.0022*
Test F-Statistic	447.0		1.2		*174.8*		*0.4*	
P-Value	*(.00)*		*(.21)*		*(.00)*		*(.99)*	

Notes: Columns 1 & 3 present f-test values for whether β(propensity) = 0, from a regression of whether a minor violation occurred during the first year post-release on officer minor violation propensity, controlling for observable characteristics about the individual. Columns 2 & 4 present joint f-test p-values from a regression of officer minor violation propensity on observable characteristics about the individual. When constructing officer minor violation propensity, Columns 3 & 4 include three agent home plan leniency types (lenient, neutral, and strict) in randomization groups U_{it}, and drop randomization groups for which there is no longer sufficient officer variation. N = 29,265 for Columns 1 and 2 and N = 22,214 for Columns 3 and 4. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Figure 2 – Variation in Officer Effects





Notes: Figures present histograms of parole officer effect measures, along with local linear regressions of the outcome of interest on the UJIVE effect measure. N = 29,265 releases to home plans in parole officer units with quasi-random officer assignment.

Table 4 – Alternative Specification – Officer Fixed Effects

	Adjusted R^2 w/o Officer Fixed Effects	% Total Variation Due to Officer	% Obs. Variation Due to Officer	% Total Variation Due to Officer
Recidivism	0.086	0.089	0%	3%
Arrest	0.064	0.066	0%	3%
PV Reincarceration	0.076	0.084	1%	11%
Minor Parole Violation	0.110	0.135	3%	23%
Sanction Type				
Written Warning	0.123	0.148	3%	20%
New Condition	0.079	0.100	2%	27%
Non-Secure Fac. Stay	0.064	0.073	1%	14%
Ever Emp. (Parole Data)	0.120	0.163	4%	36%
Ever Emp. (L&I Data)	0.097	0.100	0%	3%

Notes: Table presents adjusted R^2 measures from regressions of whether each outcome occurred during the first-year post-release on observable characteristics about the individual and randomization groups U_{it}. N = 29,265 releases to home plans in parole officer units with quasi-randomly assigned officers.

Table 5 – Relationship between Officer Effects and Survey Results

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
Survey Categories	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Interpersonal Support											
Relationship Important	3.87	.004	*.009*	.006	*.007*	-.037***	*.011*	-.007	.011	-.022**	*.011*
Care about Parolees	4.10	.006	*.007*	.012**	*.005*	.010	*.008*	-.007	.009	-.007	*.009*
Treat Nonjudgmentally	4.01	-.011	*.009*	-.021***	*.007*	-.019*	*.010*	.015	.011	.002	*.011*
Support after Mistake	4.03	.014	*.009*	-.015**	*.007*	-.024**	*.011*	-.006	.011	.001	*.011*
Learn about Parolees	3.89	-.016*	*.009*	.005	*.007*	.003	*.011*	.000	.012	-.005	*.011*
Structural Support											
Match to Suitable Interventions	3.58	.003	*.006*	.008*	*.004*	.007	*.007*	-.011	.008	.006	*.007*
Encourage Personal Improvement	3.76	.007	*.010*	.002	*.008*	.022*	*.012*	.002	.013	-.011	*.013*
Believe Parolees can Change	3.57	.004	*.007*	-.001	*.005*	.004	*.008*	-.001	.009	-.004	*.009*
Contact Support System	3.78	-.011**	*.006*	.008*	*.004*	.009	*.007*	.010	.007	.018***	*.007*
Other Categories											
Case Worker not Police Officer	3.67	-.009	*.006*	-.001	*.005*	.020**	*.008*	.006	.008	.014*	*.008*
Favorable Opinion of Par. Policies	2.45	-.003	*.004*	-.002	*.003*	-.004	*.005*	.039***	.006	.013**	*.005*
Responded	0.68	.000	*.007*	-.012**	*.005*	-.016*	*.008*	.035***	.009	-.009	*.009*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each aggregate officer survey measure, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 25,936 releases to officers who were emailed the survey, and N = 17,575 releases to officers who responded to the survey, among parole officer units with quasi-random officer assignment. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Table 6 – Aggregate Effects of Officer Demographic Characteristics

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Male Officer	0.77	-.004	*.007*	.000	*.005*	-.019**	*.008*	-.059***	*.009*	-.008	*.008*
Black Officer	0.15	-.013	*.009*	-.002	*.007*	-.014	*.011*	-.034***	*.011*	-.003	*.011*
Hispanic Officer	0.04	-.039***	*.015*	-.003	*.011*	.021	*.017*	.022	*.018*	-.011	*.018*
Officer Years of Service	9.86	.001**	*.000*	-.001***	*.000*	-.002***	*.001*	.001**	*.001*	-.001	*.001*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each officer demographic characteristics, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 29,265 releases to parole officer units with quasi-random officer assignment. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Table 7A – Outgroup Disparities

	Race Outgroup Disparities				Sex Outgroup Disparities		
	White Officer	Non-White Officer	*Overall Effect - Opposite Race*		Male Officer	Female Officer	*Overall Effect - Opposite Sex*
Recidivism							
White Parolee	0	8%	19%***	Male Parolee	0	1%	11%
Non-White Parolee	10%	0%***		Female Parolee	-23%	-33%	
Arrest							
White Parolee	0	13% *	25%***	Male Parolee	0	2%	13%
Non-White Parolee	22%	10%***		Female Parolee	-21%	-32%	
PV Reincarceration							
White Parolee	0	-3%	5%	Male Parolee	0	3%	6%
Non-White Parolee	-29%	-38%		Female Parolee	-26%	-29%	
Minor Parole Violation							
White Parolee	0	-2%	3%	Male Parolee	0	6%***	3%
Non-White Parolee	4%	-1%*		Female Parolee	-4%	-1%	
Ever Employed (Parole Data)							
White Parolee	0	-6%*	-2%	Male Parolee	0	9%***	-10%**
Non-White Parolee	-17%	-21%*		Female Parolee	-16%	4%***	
Ever Employed (L&I Data)							
White Parolee	0	-3%	-2%	Male Parolee	0	1%	-10%**
Non-White Parolee	-5%	-6%		Female Parolee	1%	11%***	

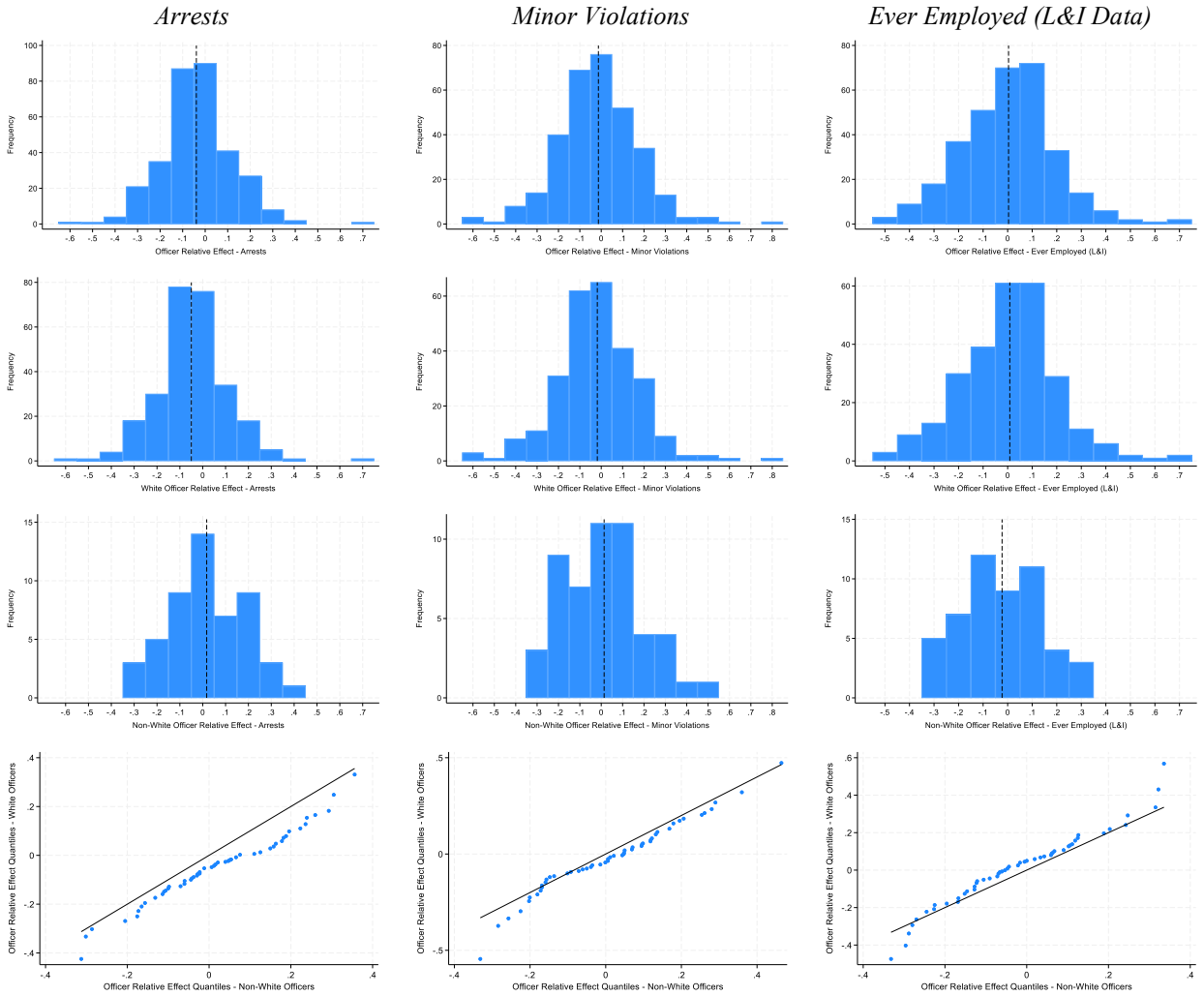
Notes: Table presents percentage point estimates of relative parolee-officer race and gender outgroup effects, from regressions of whether each outcome occurred during the first-year post-release, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 29,265 releases to parole officer units with quasi-random officer assignment. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively, for tests of whether effects differ for (1) white parolees assigned to white and nonwhite officers, (2) nonwhite parolees assigned to white and nonwhite officers, and (3) all parolees assigned to officers of the same race relative to parolees assigned to officers of the opposite race. Sex outgroup disparity statistical significance tests are constructed analogously.

Table 7B – Outgroup Estimate Details

		Recidivism		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed – L&I Data	
	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Race – Paroled ind. / Officer													
White / Non-White	0.03	.023	*.017*	.026*	*.015*	-.003	*.011*	-.006	*.018*	-.032*	*.019*	-.015	*.018*
Non-White / Non-White	0.16	-.001	*.011*	.021**	*.010*	-.036***	*.007*	-.004	*.012*	-.110***	*.013*	-.032***	*.012*
Non-White / White	0.39	.029***	*.008*	.045***	*.007*	-.028***	*.005*	.014*	*.008*	-.091***	*.009*	-.028***	*.009*
Sex – Paroled Ind. / Officer													
Male / Female	0.20	.004	*.008*	.008	*.007*	.003	*.005*	.024***	*.008*	.050***	*.009*	.004	*.009*
Female / Female	0.03	-.094***	*.018*	-.064***	*.016*	-.028**	*.012*	-.003	*.019*	.020	*.020*	.063***	*.019*
Female / Male	0.09	-.066***	*.010*	-.042***	*.009*	-.025***	*.006*	-.015	*.010*	-.084***	*.011*	.003	*.011*

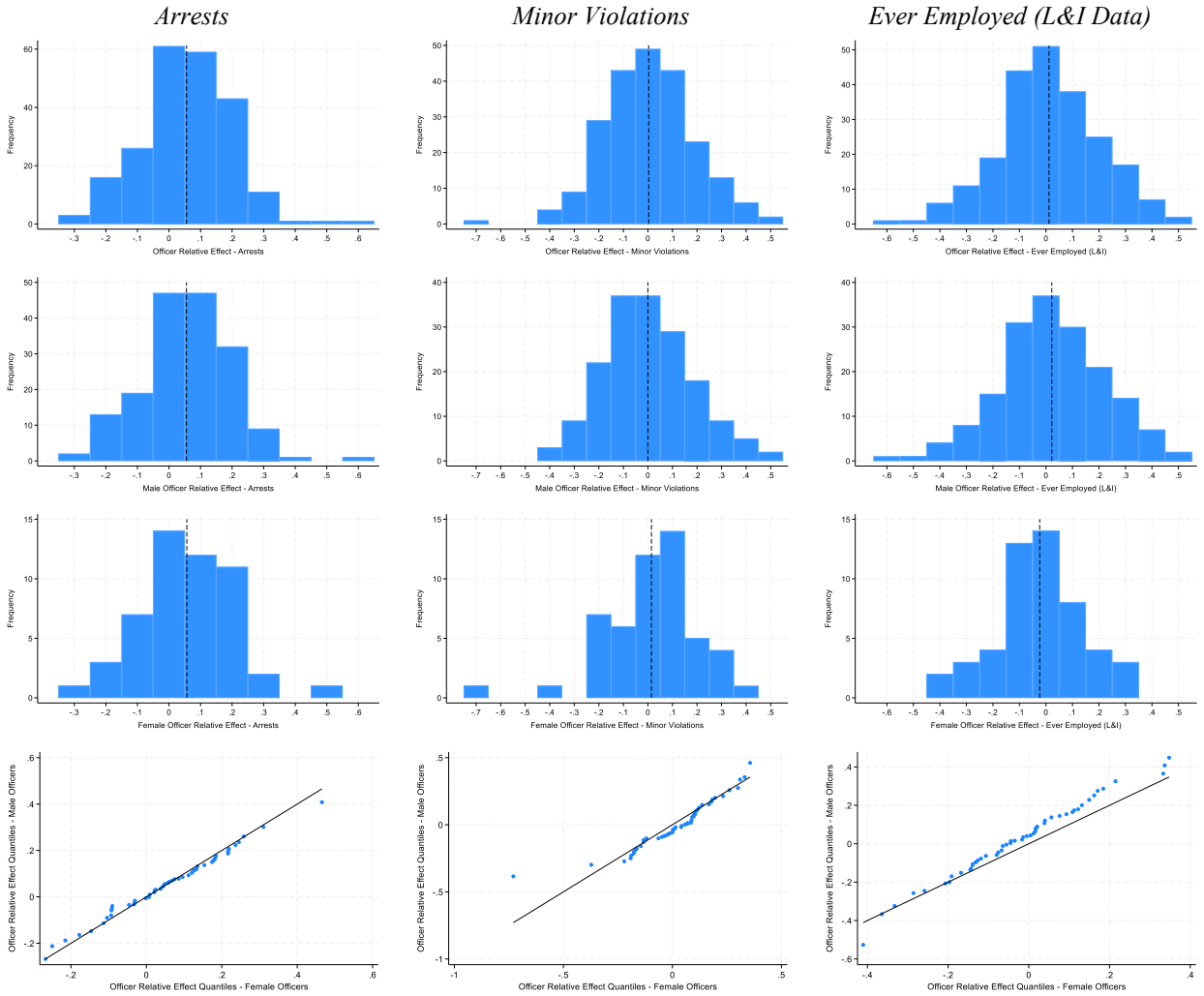
Notes: Table presents estimates used to construct percentages in Table 7A. Estimates are from regressions of whether each outcome occurred during the first-year post-release on indicators for relative parolee-officer race and gender, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 29,265 releases to parole officer units with quasi-random officer assignment. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Figure 3A – Relative Officer Effect for White vs. Non-White Paroled Individuals



Notes: The first three rows of figures present histograms of "relative officer effects" for race on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated using UJIVE when allowing officer effects to differ for white and nonwhite paroled individuals (among officers assigned at least five white and five nonwhite individuals). "Relative officer effect" is then constructed as the officer's average effect for white paroled individuals minus the officer's average effect for nonwhite paroled individuals. The first row of histograms displays this relative effect for all parole officers (N = 323), the second displays this relative effect for all white parole officers (N= 271), and the third displays this relative effect for all nonwhite parole officers (N = 52). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for white and nonwhite parole officers (for 50 quantiles).

Figure 3B – Difference in Officer Effect for Male vs. Female Paroled Individuals



Notes: The first three rows of figures present histograms of "relative officer effects" for sex on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated using UJIVE when allowing officer effects to differ for male and female paroled individuals (among officers assigned at least five male and five female individuals). "Relative officer effect" is then constructed as the officer's average effect for male paroled individuals minus the officer's average effect for female paroled individuals. The first row of histograms displays this relative effect for all parole officers (N = 228), the second displays this relative effect for all male parole officers (N= 176), and the third displays this relative effect for all female parole officers (N = 52). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for white and nonwhite parole officers (for 50 quantiles)

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Appendix A – Auxiliary Tables and Figures

Table A1 – Community Corrections Center Releases

	CCC Releases (Mean)	Home Plan Releases (Mean)
A. Demographic Characteristics		
Male	92%	88%
Black	47%	38%
Hispanic	12%	8%
Married	13%	16%
Divorced / Separated	10%	11%
No High School Degree	43%	35%
Age at Release	35.3	35.0
Violent Crime Conviction	30%	25%
Drug Crime Conviction	32%	34%
Property Crime Conviction	22%	20%
Minimum Sentence Length (Years)	3.0	2.5
Paroled at Minimum Sentence Hearing	55%	67%
Remaining Sentence at Release (Years)	3.7	3.5
Medium Recidivism Risk Level	36%	42%
High Recidivism Risk Level	43%	29%
B. Outcomes – One Year After Release		
Recidivism *(Arrested or Reincarcerated)*	39%	29%
Arrest	21%	20%
Parole Violation Reincarceration	23%	10%
Minor Parole Violation	41%	39%
Ever Employed (Parole Data)	47%	55%
Ever Employed (L&I Data)	62%	61%

Notes: This table presents summary statistics for the 50,460 home plan releases to parole (Column 1), and 50,908 Community Corrections Center releases to parole (Column 2), in Pennsylvania between 3/1/2006 and 1/1/2019.

Table A2 – Home Plan Instrument Validity

	Relevance Test		***Exogeneity Test***	
	Est.	***SE***	***Est.***	***SE***
Dependent Variable	***HP Approved***		***HP Leniency***	
Home Plan Leniency	.5011 ***	*.0247*	-	-
Male	-.0292 ***	*.0072*	.0005	*.0014*
Black	-.0807 ***	*.0058*	.0004	*.0012*
Hispanic	-.0314 ***	*.0090*	.0002	*.0018*
Married	-.0047	*.0059*	-.0006	*.0012*
Divorced / Separated	-.0149 **	*.0074*	-.0020	*.0015*
High School Graduate	.0403 ***	*.0043*	.0012	*.0009*
Violent Crime Conviction	-.0188 ***	*.0067*	.0000	*.0013*
Drug Crime Conviction	.0277 ***	*.0063*	.0000	*.0013*
Minimum Sentence Length (Years)	.0028 ***	*.0008*	-.0003 *	*.0002*
Test F-Statistic	411		0.87	
P-Value	.00		.56	

Notes: Column 1 presents f-test values for whether β(leniency) = 0, from a regression of home plan outcome on agent home plan leniency, controlling for observable characteristics about the individual. Column 2 presents joint f-test p-values from a regression of agent home plan leniency on observable characteristics about the individual. N = 43,210 initial home plan investigations in parole officer units with quasi-randomly assigned officers. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Table A3 – Robustness Check Tests for Officer Outcome Effects

	Keep Nonrandom Units	Keep Officers w Few Parolees	Keep All Observations	Let Officer Effects Vary by Year	Cond. on HP Leniency – 2 Bins	Cond. on HP Leniency – 4 Bins
A. Relevance Test	*F-Stat.*	*F-Stat.*	*F-Stat.*	*F-Stat.*	*F-Stat.*	*F-Stat.*
Recidivism (*arrested or reincar.*)	21.4	21.2	36.0	2.3	1.0	4.6
Arrest	16.9	19.8	41.4	2.2	2.5	4.7
PV Reincarceration	97.3	43.2	65.6	12.5	27.5	19.8
Minor Parole Violation (*any*)	654.3	471.2	675.8	309.0	278.6	139.7
Written Warning	719.9	522.8	789.8	318.5	321.7	183.9
New Condition	490.8	305.9	452.4	127.7	201.3	114.0
Non-Secure Facility Stay	118.0	61.0	76.4	57.8	48.6	32.0
Ever Employed (Parole Data)	1005.0	733.3	991.9	634.2	549.6	341.2
Ever Employed (L&I Data)	14.7	4.7	5.7	0.0	23.4	6.1
B. Exogeneity Test						
Recidivism (*arrested or reincar.*)	4.5	2.8	3.2	1.3	2.0	1.0
Arrest	4.4	2.9	3.8	2.5	1.6	0.7
PV Reincarceration	0.9	1.3	1.0	1.5	1.4	0.6
Minor Parole Violation (*any*)	6.9	1.3	6.8	1.7	0.4	0.6
Written Warning	6.2	1.1	6.1	1.1	0.5	0.7
New Condition	5.5	1.2	5.0	2.1	0.6	0.7
Non-Secure Facility Stay	2.1	1.5	2.5	1.6	1.4	1.4
Ever Employed (Parole Data)	3.1	2.3	2.3	1.4	2.2	2.1
Ever Employed (L&I Data)	2.7	1.8	1.8	0.6	1.9	1.5

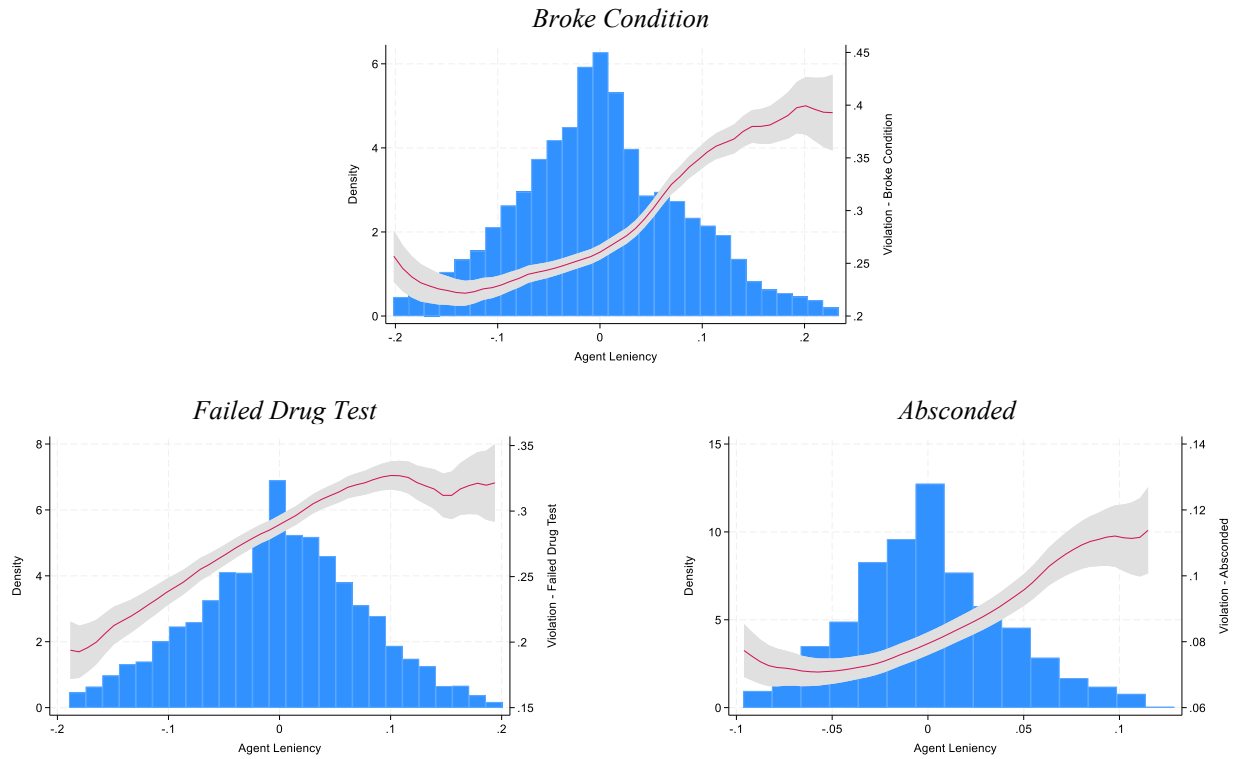
Notes: Panel A presents f-test values for whether β(propensity) = 0, from regressions of whether each outcome occurred during the first-year post-release on each officer outcome propensity measure, controlling for observable characteristics about the individual. Panel B present joint f-test values from regressions of officer outcome propensity measures on observable characteristics about the individual. Column 1 does not drop rural officer units for which officer assignment does not appear to be random (N = 38,858), Column 2 does not drop officers who are assigned less than 20 paroled individuals (N = 41,777), and Column 3 does not drop either subset of the data (N = 50,460). Column 4 presents results when officer effects are allowed to vary by year, for officer-years in which the officer was assigned at least 5 paroled individuals (N = 27,125). Columns 5 and 6 include agent home plan leniency type in randomization groups U_{it}, and drop randomization groups for which there is no longer sufficient officer variation. Column 5 splits agent home plan leniency type into two categories (lenient and strict) (N = 25,737) and Column 6 splits agent home plan leniency type into four categories (very lenient, somewhat lenient, somewhat strict, and very strict) (N = 19,660).

Table A4 – Relevance & Exogeneity Summary for CCC Releases

	Primary Outcome Propensity Measures			
	Relevance Test		***Exogeneity Test***	
	F-Stat.	*P-Val.*	*F-Stat.*	*P-Val.*
Recidivism (*arrested or reincar.*)	4.2	*(.04)*	1.2	*(.24)*
Arrest	0.5	*(.82)*	0.3	*(.99)*
PV Reincarceration	0.1	*(.74)*	2.1	*(.00)*
Minor Parole Violation	25.1	*(.00)*	1.4	*(.12)*
Sanction Type				
Written Warning	35.2	*(.00)*	1.4	*(.10)*
New Condition	31.8	*(.00)*	1.1	*(.40)*
Non-Secure Facility Stay	0.9	*(.34)*	1.7	*(.04)*
Ever Employed (Parole Data)	12.2	*(.00)*	1.6	*(.06)*
Ever Employed (L&I Data)	1.4	*(.24)*	1.1	*(.32)*

Notes: Column 1 presents f-test values for whether β(propensity) = 0, from regressions of whether each outcome occurred during the first year post-release on each agent outcome propensity measure, controlling for observable characteristics about the individual. Column 2 presents joint f-test p-values from regressions of agent outcome propensity measures on observable characteristics about the individual. N = 21,231 releases to Community Correction Centers in centers with quasi-randomly assigned officers.

Figure A1 – Variation in Officer Effects – Violation Types



Notes: Figures present histograms of parole officer effect measures, along with local linear regressions of the outcome of interest on the UJIVE effect measure. N = 29,265 releases to home plans in parole officer units with quasi-random officer assignment.

Table A5 - Relationship between Officer Effects and Individual Survey Question Responses

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
Survey Questions	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Relationship Important											
I Build Rapport	4.29	.009	*.009*	.009	*.006*	-.019*	*.010*	-.001	*.011*	.011	*.011*
Good Relationship	4.03	.005	*.008*	-.001	*.006*	-.016*	*.009*	-.010	*.009*	-.013	*.009*
Rapport Important	4.07	-.004	*.006*	.004	*.005*	.023***	*.008*	.025***	*.008*	-.008	*.008*
Parolees Comfortable	3.80	-.003	*.007*	-.005	*.005*	-.028***	*.008*	-.007	*.009*	-.008	*.008*
Parolees Talk Issues	3.15	-.003	*.005*	-.001	*.004*	-.004	*.006*	-.013*	*.007*	-.005	*.007*
Care about Parolees											
Care Parolees	3.97	-.005	*.007*	.007	*.005*	.003	*.008*	-.008	*.009*	-.005	*.008*
Want to Help	4.24	.010	*.008*	.005	*.006*	-.006	*.010*	-.013	*.010*	.010	*.010*
Treat Nonjudgmentally											
Optimistic with Parolees	3.66	.000	*.006*	-.013***	*.005*	.002	*.008*	.007	*.008*	-.005	*.008*
Praise for Success	4.23	.001	*.008*	-.013**	*.006*	-.011	*.009*	.000	*.010*	.013	*.010*
Nonjudgemental	4.14	-.007	*.007*	-.003	*.005*	-.015*	*.008*	.001	*.008*	-.007	*.008*
Support after Mistake											
Provide Warning	3.99	.006	*.006*	-.005	*.005*	-.013*	*.008*	-.015*	*.008*	-.005	*.008*
Work with Parolees	3.86	.002	*.008*	.000	*.006*	-.004	*.009*	.003	*.009*	.006	*.009*
Explain Consequences	4.24	.006	*.007*	-.007	*.006*	.001	*.009*	-.004	*.009*	.003	*.009*
Learn about Parolees											
Know Parolee Situation	4.18	-.003	*.007*	.005	*.005*	-.008	*.008*	-.016*	*.009*	.002	*.008*
Consider Parolee Views	3.76	-.005	*.008*	-.003	*.006*	.014	*.009*	.029***	*.010*	-.011	*.010*
Think about Reason Behavior	3.74	-.004	*.006*	.000	*.005*	.002	*.008*	-.005	*.008*	.005	*.008*
Match to Suitable Interventions											
Programming Match Risk	3.54	.008	*.006*	-.003	*.004*	.002	*.007*	-.009	*.007*	.005	*.007*
Interventions Match Risk	3.29	-.003	*.006*	.012***	*.004*	-.001	*.007*	-.011	*.007*	.008	*.007*
Identify Criminogenic Needs	3.90	.000	*.006*	-.004	*.005*	.002	*.008*	-.020***	*.008*	-.001	*.008*
Encourage Personal Improvement											
Identify Triggers	3.69	.006	*.006*	.009**	*.005*	.017**	*.007*	.009	*.008*	-.002	*.008*
Help Success Society	4.11	.002	*.008*	-.005	*.006*	.000	*.009*	-.008	*.010*	-.016	*.010*
Skill Building Discussions	3.07	.011*	*.006*	.006	*.004*	-.004	*.007*	-.008	*.007*	.006	*.007*
Set Parolee Goals	3.71	-.007	*.007*	-.006	*.006*	.005	*.009*	.016*	*.009*	-.017*	*.009*
Demonstrate Prosocial Skills	4.22	-.001	*.006*	-.004	*.004*	.018**	*.007*	.014*	*.007*	-.001	*.007*
Promote Prosocial	4.05	.003	*.008*	.002	*.006*	.003	*.009*	.014	*.009*	.009	*.009*
Mentor Parolees	3.50	-.002	*.005*	-.002	*.004*	-.014**	*.006*	-.010	*.006*	.000	*.006*
Believe Parolees can Change											
Parolees Can Change	3.51	-.003	*.005*	.004	*.004*	-.005	*.006*	.003	*.007*	.008	*.007*
Parolees Can Learn	3.68	.002	*.006*	-.006	*.005*	.009	*.007*	-.005	*.008*	-.010	*.007*
Parolees Initially Motivated	3.52	.003	*.005*	-.003	*.004*	-.002	*.006*	.005	*.006*	.000	*.006*
Contact Support System											
Contact Support System	3.78	-.019***	*.007*	.011**	*.005*	.008	*.008*	-.005	*.008*	.020**	*.008*

		Arrested		PV Reincar.		Minor PV		Employed - Parole Data		Employed - L&I Data	
Survey Questions	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
Case Worker not Police Officer	3.31	-.004	*.004*	-.001	*.003*	.005	*.004*	.006	*.004*	.005	*.004*
Case Worker	3.75	-.007	*.005*	.004	*.003*	.023***	*.005*	.013**	*.006*	.008	*.006*
Supervision Balance	3.95	-.002	*.008*	.000	*.006*	.008	*.009*	.023**	*.010*	-.008	*.010*
Reentry Important											
Favorable Opinion of Par. Policies	2.90	.000	*.004*	-.007**	*.003*	-.010*	*.005*	.016***	*.006*	.008	*.005*
Current Policies Supportive	2.00	-.004	*.005*	.004	*.004*	.009	*.006*	.028***	*.006*	.007	*.006*
Workload Achievable											
Parolees Hopeful at Max Date	3.47	.001	*.006*	.005	*.005*	.015	*.008*	-.012	*.008*	.006	*.008*
Parolees Seem Hopeful	3.31	-.004	*.004*	-.001	*.003*	.005	*.004*	.006	*.004*	.005	*.004*

Notes: Table presents estimates from regressions of whether each outcome occurred during the first-year post-release on each officer survey response, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 25,936 releases to officers who were emailed the survey, and N = 17,575 releases to officers who responded to the survey, among parole officer units with quasi-random officer assignment. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Table A6 – Leniency Measure Monotonicity by Race and Sex

		Recidivism		**Arrested**		**PV Reincar.**		**Minor PV**		**Employed - Parole Data**		**Employed – L&I Data**	
	Mean	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE	Est.	SE
A. Within-Subsample Monotonicity													
White Paroled Individuals	0.45	.131***	*.049*	.096*	*.050*	.275***	*.050*	.465***	*.039*	.577***	*.033*	*.153****	*.050*
Non-White Paroled Individuals	0.55	.034	*.051*	.050	*.051*	.170***	*.044*	.569***	*.034*	.666***	*.030*	*.107***	*.050*
Male Paroled Individuals	0.89	.091**	*.036*	.096**	*.037*	.213***	*.035*	.538***	*.026*	.626***	*.023*	*.121****	*.037*
Female Parole Individuals	0.11	.112	*.134*	.041	*.134*	.399***	*.112*	.546***	*.093*	.659***	*.083*	.150	*.124*
B. Across-Subsample Monotonicity													
White Paroled Individuals	0.45	-.025	*.017*	-.002	*.010*	.002	*.017*	.088***	*.014*	.101***	*.014*	*.005*	*.018*
Non-White Paroled Individuals	0.55	.018	*.015*	.018	*.014*	.021	*.015*	.060***	*.014*	.079***	*.013*	*.005*	*.017*
Male Paroled Individuals	0.89	-.005	*.007*	-.009	*.007*	.023***	*.009*	.042***	*.008*	.048***	*.007*	.015*	*.008*
Female Parole Individuals	0.11	.076	*.130*	.040	*.129*	.387***	*.105*	.588***	*.093*	.666***	*.086*	*.169*	*.127*

Notes: Table presents estimates from regressions of each outcome on the associated agent outcome propensity measure, controlling for observable characteristics about the individual, for different race and sex subsamples. Panel A presents estimates when propensity measures are created using the full set of data, but regressions are run using individual subsets of interest. Panel B presents estimates when propensity measures are created using the full set of data except for the subsample of interest, and regressions are run using only the subset of interest. N = 29,265 releases to home plans in parole officer units with quasi-randomly assigned officers. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively.

Table A7 – Officer Outgroup Effect Robustness Checks

A. Keep Officers with Few Assigned Paroled Individual

	Racial Outgroup Disparities				**Sex Outgroup Disparities**		
	White Officer	Non-White Officer	*Overall Effect - Opposite Race*		Male Officer	Female Officer	*Overall Effect - Opposite Sex*
Recidivism							
White Parolee	0	3%	12%**	Male Parolee	0	-1%	10%*
Non-White Parolee	9%	0%***		Female Parolee	-22%	-32%**	
Arrest							
White Parolee	0	7%	17%**	Male Parolee	0	-1%	13%**
Non-White Parolee	20%	10%***		Female Parolee	-23%	-37%**	
PV Reincarceration							
White Parolee	0	-6%	3%	Male Parolee	0	4%	7%
Non-White Parolee	-24%	-32%		Female Parolee	-18%	-21%	
Minor Parole Violation							
White Parolee	0	1%	7%	Male Parolee	0	8%***	3%
Non-White Parolee	4%	-2%***		Female Parolee	-6%	-1%	
Ever Employed (Parole Data)							
White Parolee	0	-7%***	-3%	Male Parolee	0	10%***	-6%*
Non-White Parolee	-16%	-21%***		Female Parolee	-17%	-2%***	
Ever Employed (L&I Data)							
White Parolee	0	-5%*	-4%	Male Parolee	0	2%	-6%**
Non-White Parolee	-6%	-7%		Female Parolee	0%	8%***	

Notes: Table presents percentage point estimates of relative parolee-officer race and gender outgroup effects, from regressions of whether each outcome occurred during the first-year post-release, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 41,777 releases to parole officer units with quasi-random officer assignment, without dropping officers assigned less than 20 paroled individuals. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively, for tests of whether effects differ for (1) white parolees assigned to white and nonwhite officers, (2) nonwhite parolees assigned to white and nonwhite officers, and (3) all parolees assigned to officers of the same race relative to parolees assigned to officers of the opposite race. Sex outgroup disparity statistical significance tests are constructed analogously.

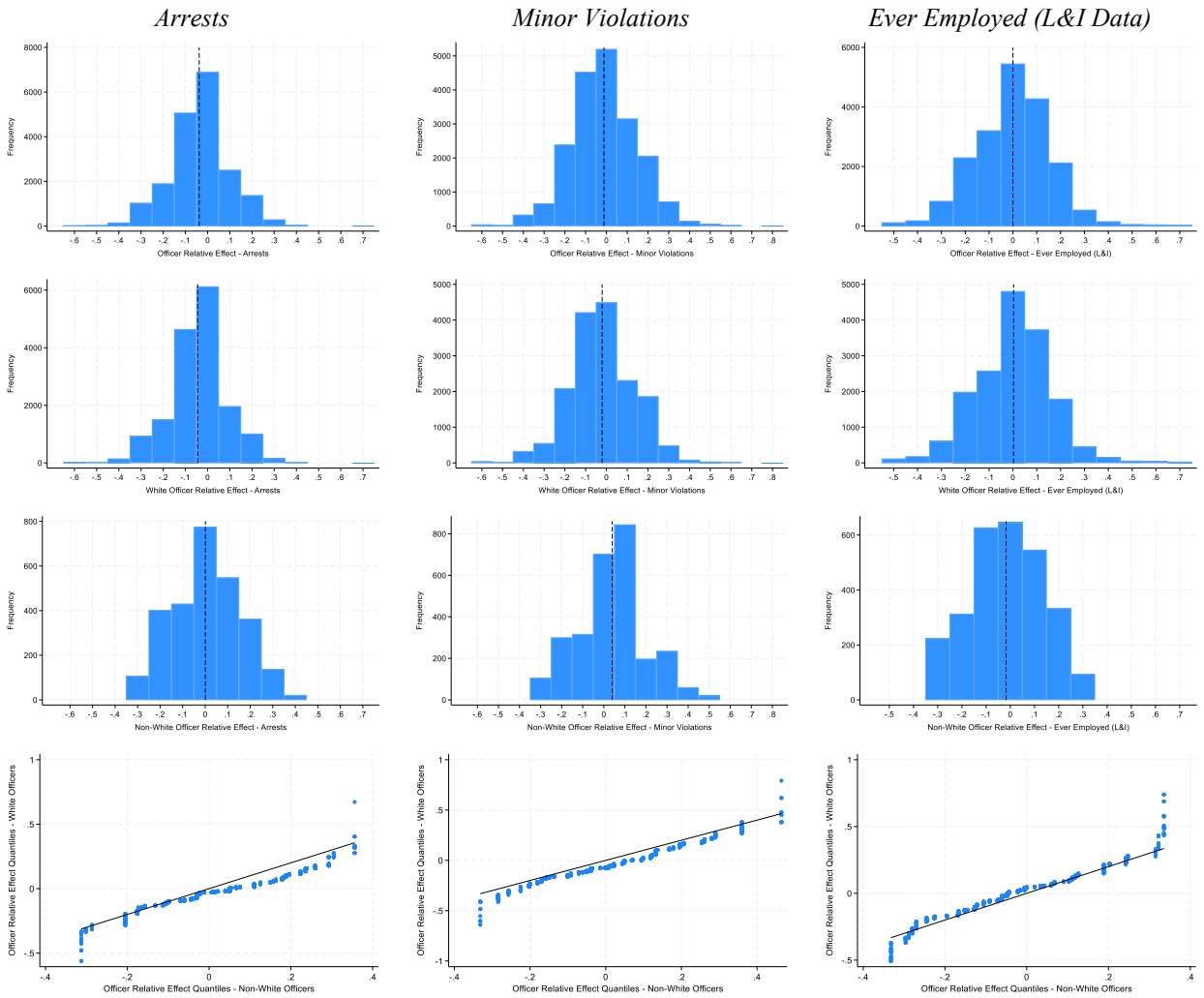
B. Keep All Observations

	Racial Outgroup Disparities				Sex Outgroup Disparities		
	White Officer	Non-White Officer	*Overall Effect - Opposite Race*		Male Officer	Female Officer	*Overall Effect - Opposite Sex*
Recidivism							
White Parolee	0	0%	8%	Male Parolee	0	-1%	10%*
Non-White Parolee	10%	2%***		Female Parolee	-22%	-32%**	
Arrest							
White Parolee	0	4%	13%**	Male Parolee	0	-1%	13%**
Non-White Parolee	22%	13%**		Female Parolee	-23%	-37%**	
PV Reincarceration							
White Parolee	0	-13%	-5%	Male Parolee	0	4%	7%
Non-White Parolee	-24%	-31%		Female Parolee	-18%	-21%	
Minor Parole Violation							
White Parolee	0	1%	9%**	Male Parolee	0	8%***	3%
Non-White Parolee	9%	1%***		Female Parolee	-6%	-1%	
Ever Employed (Parole Data)							
White Parolee	0	-10%***	-4%	Male Parolee	0	10%***	-6%*
Non-White Parolee	-15%	-21%***		Female Parolee	-17%	-2%***	
Ever Employed (L&I Data)							
White Parolee	0	-2%	-1%	Male Parolee	0	2%	-6%**
Non-White Parolee	-4%	-5%		Female Parolee	0%	8%***	

Notes: Table presents percentage point estimates of relative parolee-officer race and gender outgroup effects, from regressions of whether each outcome occurred during the first-year post-release, controlling for observable characteristics about the individual and randomization groups U_{it}. N = 50,460 releases to home plans. *, **, and *** denote statistical significance at the .10, .05, and .01 levels, respectively, for tests of whether effects differ for (1) white parolees assigned to white and nonwhite officers, (2) nonwhite parolees assigned to white and nonwhite officers, and (3) all parolees assigned to officers of the same race relative to parolees assigned to officers of the opposite race. Sex outgroup disparity statistical significance tests are constructed analogously.

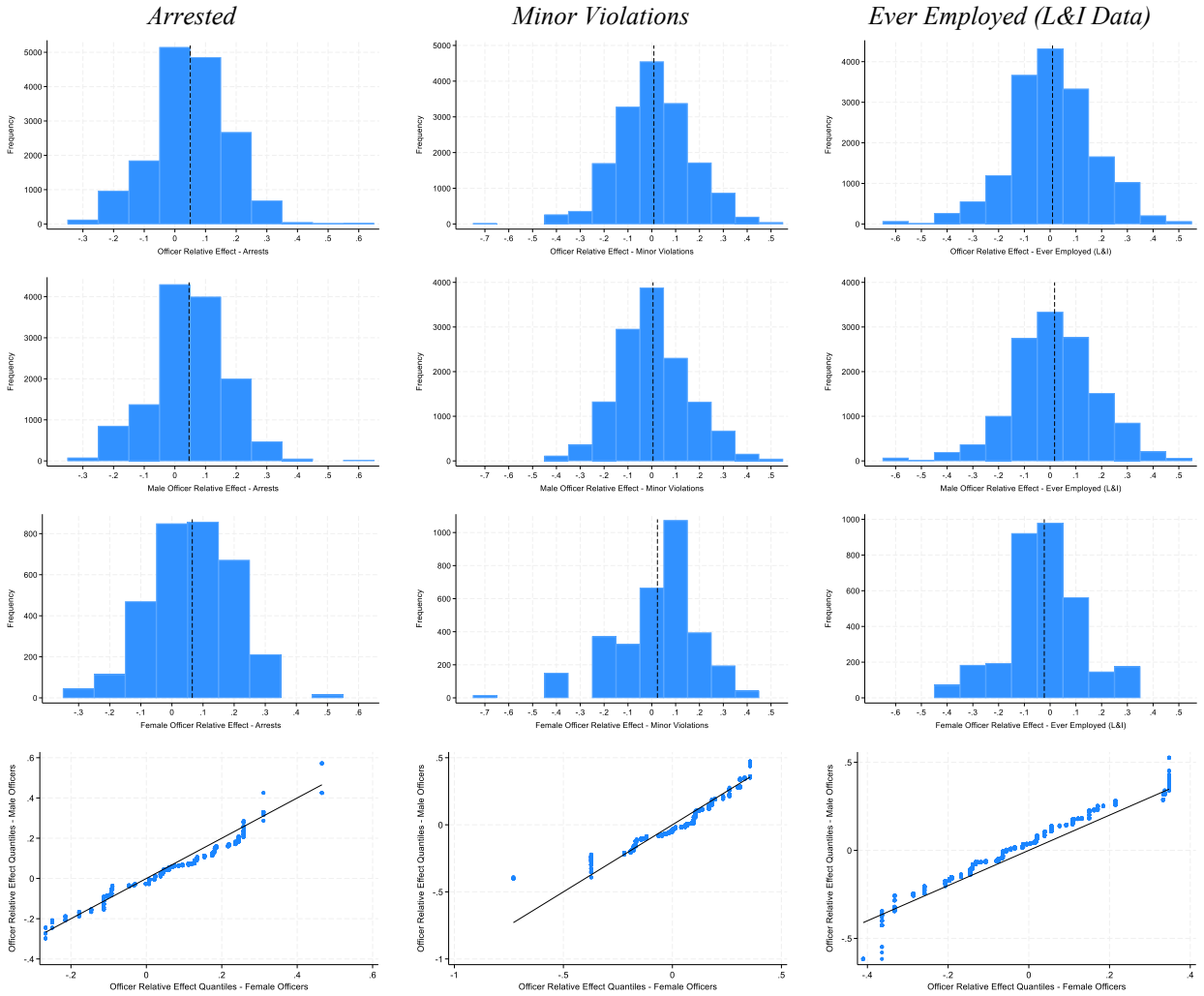
Figure A2 – Relative Officer Effect Robustness Checks

A. Relative Effect for White vs. Nonwhite Paroled Individuals – Paroled Individual-Level Data



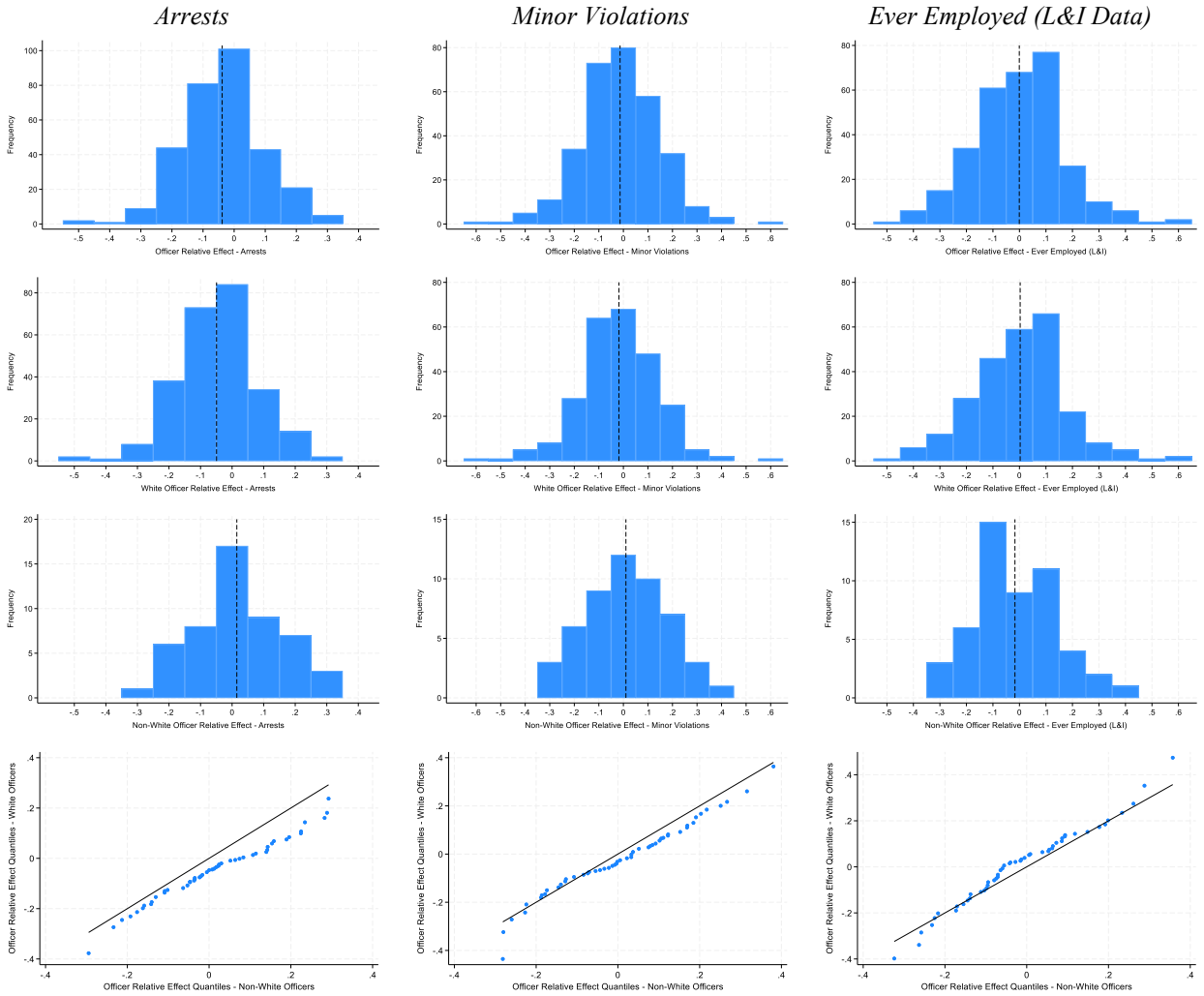
Notes: The first three rows of figures present histograms of "relative officer effects" for race on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated using UJIVE when allowing officer effects to differ for white and nonwhite paroled individuals (among officers assigned at least five white and five nonwhite individuals). "Relative officer effect" is then constructed as the officer's average effect for white paroled individuals minus the officer's average effect for nonwhite paroled individuals. The first row of histograms displays this relative parole officer effect for each paroled individual (N = 19,361), the second displays this relative effect for individuals assigned to white parole officers (N= 16,583), and the third displays this relative effect for individuals assigned to nonwhite parole officers (N = 2,798). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for individuals assigned to white and nonwhite parole officers (for 50 quantiles).

B. Relative Effect for Male vs. Female Paroled Individuals – Paroled Individual-Level Data



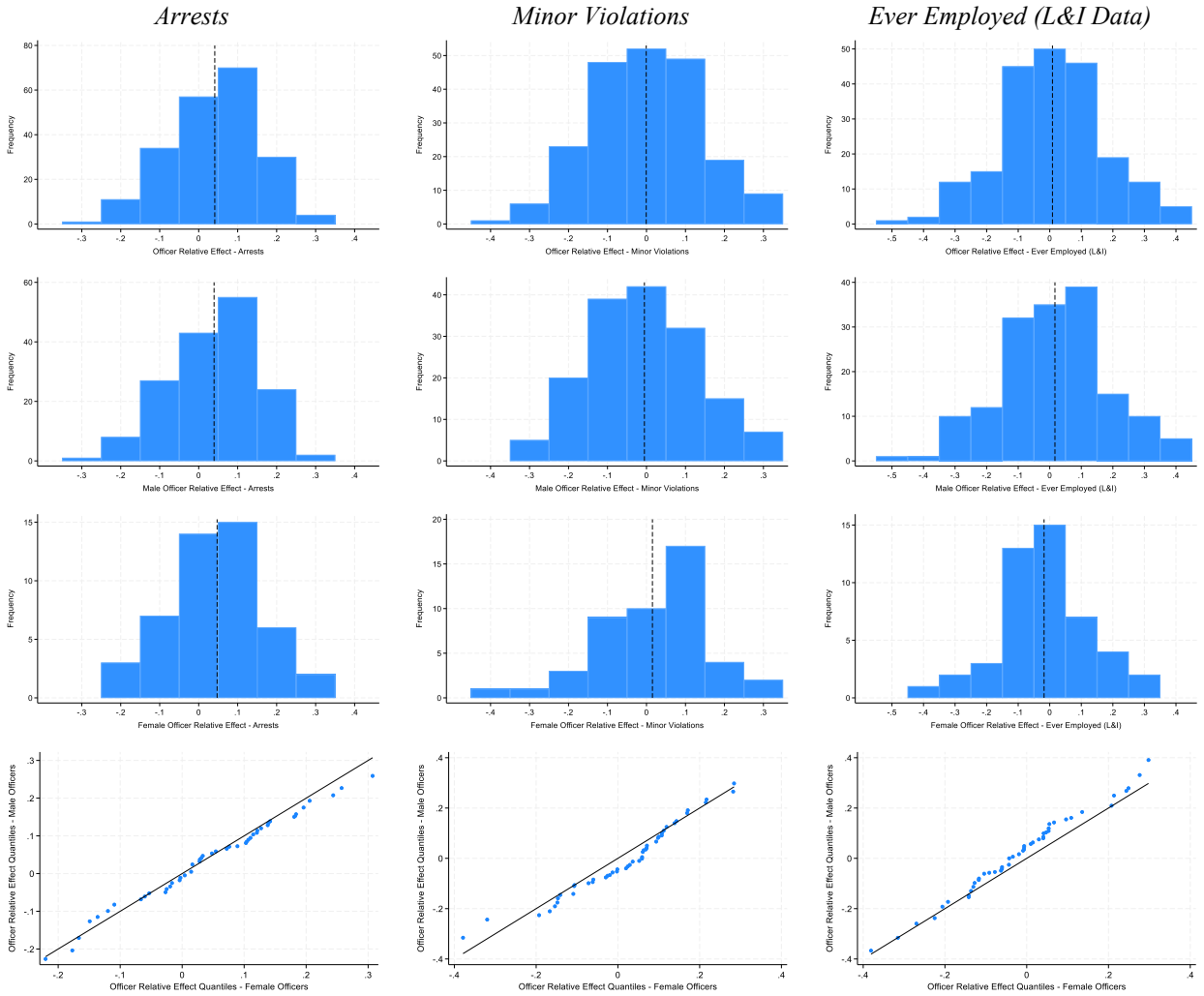
Notes: The first three rows of figures present histograms of "relative officer effects" for sex on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated using UJIVE when allowing officer effects to differ for male and female paroled individuals (among officers assigned at least five male and five female individuals). "Relative officer effect" is then constructed as the officer's average effect for male paroled individuals minus the officer's average effect for female paroled individuals. The first row of histograms displays this relative parole officer effect for each paroled individual (N = 16,437), the second displays this relative effect for individuals assigned to male parole officers (N= 13,204), and the third displays this relative effect for individuals assigned to female parole officers (N = 3,233). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for individuals assigned to male and female parole officers (for 50 quantiles).

C. Relative Effect for White vs. Nonwhite Paroled Individuals – Inverse Propensity Weights



Notes: The first three rows of figures present histograms of "relative officer effects" for race on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated as average residuals for each parolee race group, from a regression of the outcome on randomization groups U_{it} and using inverse propensity weights to balance the number of individuals of each race assigned to each officer (among officers assigned at least five white and five nonwhite individuals). "Relative officer effect" is then constructed as the officer's average effect for white paroled individuals minus the officer's average effect for nonwhite paroled individuals. The first row of histograms displays this relative parole officer effect for each officer (N = 320), the second displays this relative effect for white officers (N= 268), and the third displays this relative effect for nonwhite officers (N = 52). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for white and nonwhite parole officers (for 50 quantiles).

D. Relative Effect for Male vs. Female Paroled Individuals – Inverse Propensity Weights



Notes: The first three rows of figures present histograms of "relative officer effects" for sex on arrests, minor parole violations, and tax record-documented employment. Officer effects on each outcome are first estimated as average residuals for each parolee sex group, from a regression of outcome on randomization groups U_{it} and using inverse propensity weights to balance the number of individuals of each sex assigned to each officer (among officers assigned at least five male and five female individuals). "Relative officer effect" is then constructed as the officer's average effect for male paroled individuals minus the officer's average effect for female paroled individuals. The first row of histograms displays this relative parole officer effect for each officer (N = 227), the second displays this relative effect for male officers (N= 176), and the third displays this relative effect for female officers (N = 51). The vertical dotted lines denote the mean of each distribution. The fourth row of figures presents quantile-quantile plots that compare the values of the effect measure distributions for male and female parole officers (for 50 quantiles).

Appendix B - Parole Officer Survey Questions

"In my opinion…"

Possible responses - Strongly Agree, Agree, No Opinion, Disagree, Strongly Disagree

1. Promoting pro-social behavior is an important part of my job as a parole officer/parole supervisor.
2. Building rapport with parolees is important to their success under supervision.
3. Knowing each parolee's current situation makes my job easier.
4. Having a good relationship with a parolee makes my job easier.
5. Current Parole Board policies are supportive of offender rehabilitation.
6. One of my roles as a parole officer/parole supervisor is to help parolees become successful members of society.
7. One of my roles as a parole officer/parole supervisor is to mentor parolees.
8. Parolees are unable to change their negative behaviors.
9. Parolees are capable of learning prosocial ways of thinking.
10. The workload demands set forth by Central Office are achievable.
11. Select "Strongly agree" here.
12. I consider myself more of a case worker than a police officer.
13. Supervision should be balanced between social work and law enforcement.
14. Reentry is part of supervision.

"Consider your interactions with parolees and choose the most representative response:…"

Possible responses - Almost Always, Frequently, Sometimes, Occasionally, Hardly Ever

15. I care about parolees as people.
16. I am optimistic with parolees.
17. Parolees seem to feel comfortable enough to be open and honest with me.
18. I praise parolees for the good things they do.
19. I want to help parolees.
20. I consider parolees' views.
21. Parolees seem to keep important issues to themselves and won't tell me about them.
22. I think about the underlying reasons for parolee behavior.
23. When a parolee makes a mistake, I work with him to find out what brought him to that point.
24. I use a nonjudgmental approach in my interactions with parolees.
25. I match programming to a parolee's risk level.
26. I help parolees identify triggers and develop plans to effectively address those triggers.
27. I have frequent contact with parolees' support systems.
28. I demonstrate prosocial skills before asking a parolee to do so.
29. I am able to use risk level to apply appropriate interventions to each parolee.
30. I build rapport with the parolees on my caseload.
31. When parolees are going in a bad direction, I talk with them before taking serious action.
32. If a parolee breaks the rules, I calmly explain the consequences of his actions.

33. At the first contact, right after release from prison, parolees seem motivated to change.
34. Prior to maxing out of supervision, parolees seem hopeful about their future.
35. I include skill building discussions in each meeting with parolees.
36. I work with parolees to set goals for their personal improvement.
37. I can identify a parolee's criminogenic needs.